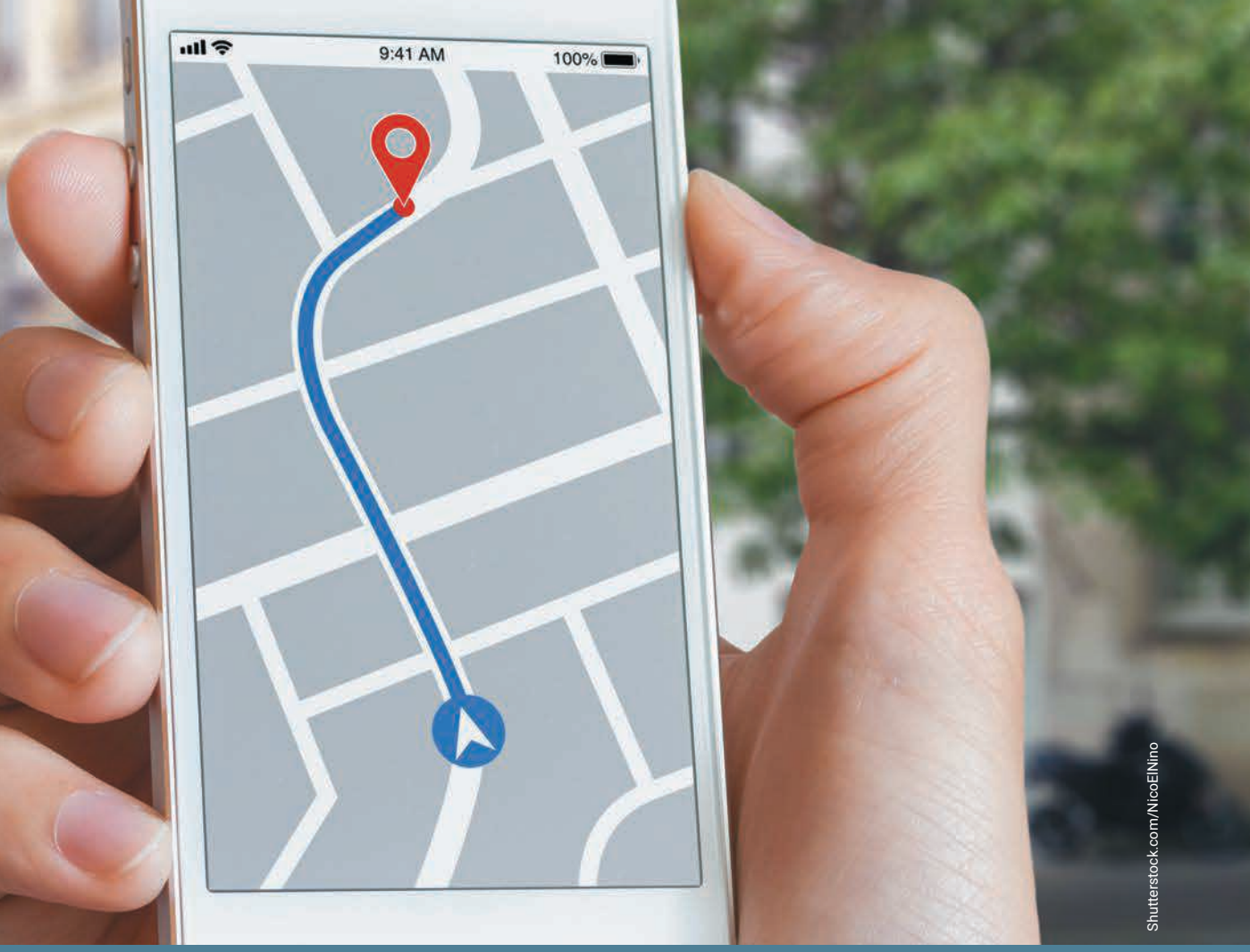


11

NUMBER AND ALGEBRA

COORDINATE GEOMETRY AND GRAPHS

The use of a coordinate system allows us to monitor the location of objects on Earth or in space. Whether reading street maps, using global positioning systems or managing road traffic flow, coordinates provide an accurate method for indicating position. Coordinates have important applications in many areas including astronomy, architecture, engineering and art design.



Chapter outline

	Working mathematically				
11.01 The length of an interval	U	F		R	C
11.02 The midpoint of an interval	U	F		R	C
11.03 The gradient of a line	U	F		R	C
11.04 Graphing linear equations	U	F		R	C
11.05 The gradient-intercept equation $y = mx + c$	U	F			
11.06 Finding the equation of a line	U	F			
11.07 Solving linear equations graphically	U	F			
11.08 Applying linear equations	U	F		R	C
11.09 Direct variation*	U	F	PS	R	
11.10 Graphing quadratic equations $y = ax^2 + c$	U	F		R	C
11.11 Graphing quadratic equations $y = ax^2 + bx + c$*	U	F		R	C
11.12 Applying quadratic equations	U	F		R	C
11.13 Graphing circles*	U	F			

***PATH**

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Wordbank

direct variation A relationship between 2 variables in which one variable is a constant multiple of the other, for example, if $y = 8.5x$, then y is directly proportional to x

gradient The steepness of a line or interval, measured by the fraction $\frac{\text{rise}}{\text{run}}$

interval A part of a line with a definite length

linear equation An equation whose graph is a straight line

midpoint The point in the middle of an interval or halfway between 2 given points

parabola A U-shaped curve that is the graph of a quadratic equation

quadratic equation An equation involving a variable squared (power of 2), such as $y = 3x^2 - 6$, whose graph is a curve called a parabola

y-intercept The y-value at which a graph cuts the y-axis

Quiz
Wordbank 11

Videos (13):

11.01 The length of an interval • Distance, midpoint and gradient formulas

11.02 Distance, midpoint and gradient formulas

11.03 The gradient of a line • Gradient (slope) of a line • Distance, midpoint and gradient formulas

11.04 Graphing linear equations • Testing if a point lies on a line • Horizontal and vertical lines

11.05 Graphing $y = mx + c$

11.06 Finding the equation of a line

11.08 Applying linear functions

11.09 Direct proportion

11.10 The parabola $y = ax^2 + c$

11.12 Applying quadratic functions

Twig video (1):

SkillCheck Coordinate geometry: Descartes

PhET interactives (5):

11.04 Graphing lines • Function builder: Basics • Function builder

11.05 Graphing slope-intercept

11.10 Graphing quadratics

Quizzes (5):

- Wordbank 11
- SkillCheck 11
- Mental skills 11
- Language of maths 11
- Test yourself 11

Skillsheet (1):

SkillCheck The number plane

Worksheets (17):

SkillCheck Tables of values

11.01 Length of an interval

11.03 Drawing gradients • A page of intervals • A page of lines • Gradient between two points • Gradient, midpoint and distance

11.04 Graphing linear equations • A page of intervals • A page of lines • A page of number planes • Number plane grid paper • Straight-line equations

11.06 Finding the equation of a line • A page of intervals • A page of lines

11.10 Graphing parabolas

11.11 Graphing quadratic functions • Graphing quadratics • Features of a parabola

Mind map Coordinate geometry and graphs (Advanced)

Puzzles (7):

11.01 Length of a line segment

11.03 Find the gradient 1 • Intervals match-up

11.05 Equation of a line

11.06 Straight-line equations

11.13 Finding coordinates for given segment lengths

Language of maths Coordinate geometry crossword

Technology (3):

11.02 Finding midpoints • Midpoint and distance between 2 points

11.10, 11.13 Curve sketcher

Spreadsheets (4):

11.02 Finding midpoints • Midpoint and distance between 2 points

11.05 Drawing straight lines: $y = mx + c$

11.10, 11.13 Curve sketcher

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- find the distance between 2 points on a number plane
- find the midpoint and gradient of an interval on a number plane
- (PATH) use the distance, midpoint and gradient formulas for an interval on the number plane
- graph a linear equation on the number plane
- test whether a point lies on a line
- use the gradient–intercept equation of a straight line $y = mx + c$
- find the equation of a line from its graph
- solve linear equations graphically
- solve problems that can be modelled by linear equations
- (PATH) graph and solve problems involving direct variation
- graph quadratic equations of the form $y = ax^2$ and $y = ax^2 + c$
- (PATH) graph quadratic equations of the form $y = ax^2 + bx + c$
- solve problems that can be modelled by quadratic equations
- (PATH) graph circles with equations of the form $x^2 + y^2 = r^2$

SkillCheck ANSWERS ON P. 644

1 Complete each table of values using the given equation.

a $y = -2x + 1$

x	−2	−1	0	1	2
y					

b $y = \frac{x}{2} + 1$

x	−2	−1	0	1	2
y					

2 Given the equation $y = 5 - 2x$, find:

a y when $x = 0$ **b** y when $x = 3$

c x when $y = 0$ **d** x when $y = 7$



Quiz
SkillCheck 11

Worksheet
Tables of values

Skillsheet
The number plane



Video
Coordinate geometry: Descartes

11.01 The length of an interval

Worksheet
Length of an interval

Puzzle
Length of a line segment

A line is usually drawn with arrows at both ends to show that it runs infinitely in both directions. However, an **interval** or line segment is part of a line, with endpoints on both sides and a definite length. We can calculate the length of an interval if we know the coordinates of its endpoints.

Example 1

Find the distance between the points:

a $P(-4, -2)$ and $Q(2, 6)$

b $A(-2, 5)$ and $B(2, -2)$

Solution

a With P and Q plotted on a number plane, construct a right-angled triangle PTQ such that PQ is the hypotenuse.

$PT = 6$ units, $TQ = 8$ units.

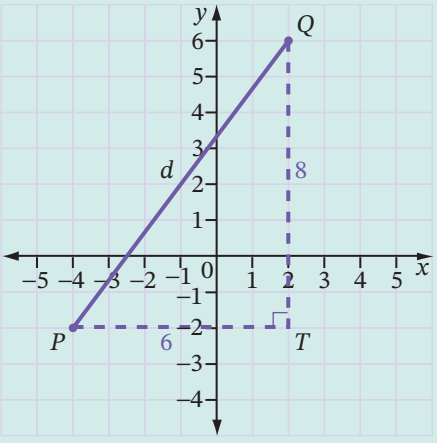
Let the length of PQ be d, so by Pythagoras' theorem:

$d^2 = 6^2 + 8^2$

$= 100$

$d = \sqrt{100} = 10$

The distance between the points P and Q is 10 units.



b By plotting A and B and constructing $\triangle ABC$, we have:

$AB^2 = 7^2 + 4^2$

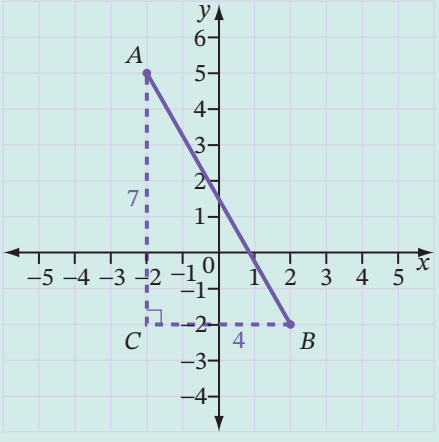
$= 65$

$AB = \sqrt{65}$

$= 8.0622...$

≈ 8.06 units

We can write the distance AB in **surd** form, $\sqrt{65}$ units as an exact answer, or we can approximate (round) it as a decimal, 8.06 units.



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Example 2

a Plot the points $A(0, 6)$, $B(5, 6)$, $C(5, 2)$ and $D(-4, 2)$ on a number plane and join them to make the quadrilateral $ABCD$.

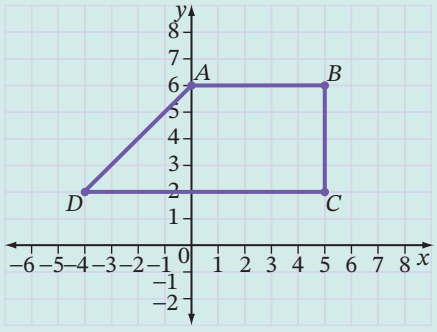
b What type of quadrilateral is $ABCD$?

c Find the exact length of AD.

d Hence find the perimeter of $ABCD$, correct to 2 decimal places.

Solution

a



b $ABCD$ is a trapezium.

c $AD^2 = 4^2 + 4^2$ By Pythagoras' theorem.

$= 32$

$AD = \sqrt{32}$ units In exact surd form.

d By counting grid squares, $AB = 5$, $BC = 4$, $CD = 9$.

Perimeter of $ABCD = 5 + 4 + 9 + \sqrt{32}$

$= 23.656...$

≈ 23.66 units

The distance formula

PATH

The method for finding the length of an interval can be summarised by a formula.

The is called the **distance formula** because it is used to calculate the distance (d) between any 2 points $P(x_1, y_1)$ and $Q(x_2, y_2)$, in other words, the length of the interval PQ.

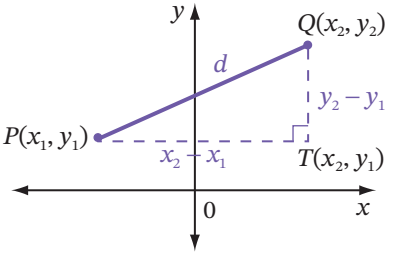
Length of $PT = x_2 - x_1$

Length of $TQ = y_2 - y_1$

By Pythagoras' theorem:

$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$\therefore d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$



PATH

The distance formula

The distance (d) between the 2 points (x_1, y_1) and (x_2, y_2) is given by the formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Group the x-coordinates together, group the y-coordinates together.

Videos
The length of an interval

Distance, midpoint and gradient formulas

Example 3

Find the length of the interval joining the points:

a $A(2, 6)$ and $B(4, 7)$, correct to one decimal place

b $P(-3, 5)$ and $Q(9, -11)$

Solution

a For $A(2, 6)$ and $B(4, 7)$:

$x_1 = 2, y_1 = 6, x_2 = 4, y_2 = 7$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(4-2)^2 + (7-6)^2}$$
$$= \sqrt{2^2 + 1^2}$$
$$= \sqrt{5}$$
$$= 2.2360...$$
$$\approx 2.2 \text{ units}$$

Group the x-coordinates together, group the y-coordinates together.

b For $P(-3, 5)$ and $Q(9, -11)$:

$x_1 = -3, y_1 = 5, x_2 = 9, y_2 = -11$

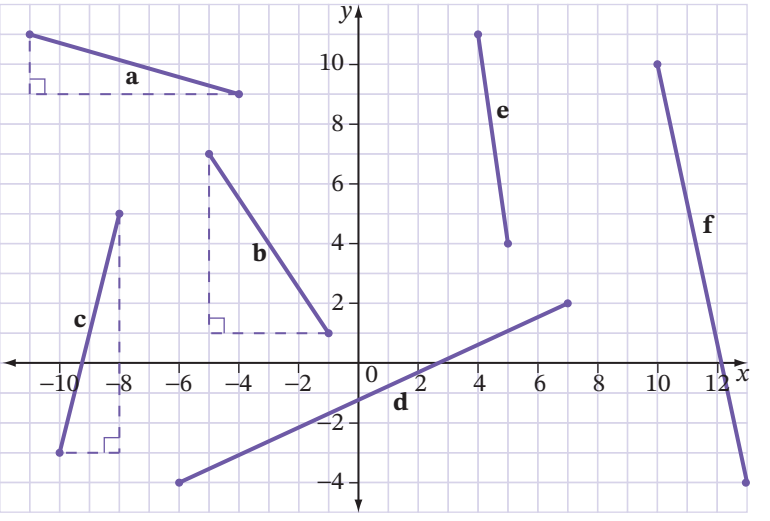
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{[9-(-3)]^2 + (-11-5)^2}$$
$$= \sqrt{12^2 + (-16)^2}$$
$$= \sqrt{400}$$
$$= 20 \text{ units}$$

EXERCISE 11.01 ANSWERS ON P. 644

The length of an interval U F R C

EXAMPLE 1

1 Calculate the length of each interval shown on this number plane. Express your answers correct to one decimal place, where necessary.



Foundation Standard Complex

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2 Plot each pair of points on a number plane and construct a right-angled triangle to find the distance between the 2 points. Express your answer:

i as a surd **ii** correct to one decimal place.

a $A(7, 6)$ and $B(1, 9)$ **b** $P(0, -5)$ and $Q(6, 4)$

c $G(9, -2)$ and $H(1, 6)$ **d** $W(-5, -4)$ and $X(-3, 4)$

3 Use the distance formula to check your answers to question **2**.

4 What is the exact distance between the points $Y(6, -3)$ and $Z(-2, 4)$? Select **A**, **B**, **C** or **D**.

A $\sqrt{113}$ units **B** $\sqrt{65}$ units **C** $\sqrt{57}$ units **D** $\sqrt{15}$ units

5 A triangle is formed by joining the points $A(5, 2)$, $B(-2, 3)$ and $C(1, 7)$. Find, correct to 3 decimal places:

a the length AB **b** the length BC

c the length AC **d** the perimeter of $\triangle ABC$

6 The vertices of triangle RST are $R(-1, -1)$, $S(1, 3)$ and $T(3, 1)$.

a Find the exact length of each side of the triangle.

b What type of triangle is $\triangle ABC$? R C

7 A quadrilateral is formed by joining the points $M(-3, 5)$, $N(5, 5)$, $P(8, 1)$ and $T(-3, 1)$.

a What type of quadrilateral is $MNPT$? R C

b Find the perimeter of $MNPT$.

c Find the area of $MNPT$.

8 The vertices of a quadrilateral are $H(1, 6)$, $I(7, 2)$, $J(3, -4)$ and $K(-3, 0)$.

a What type of quadrilateral is $HIJK$? R C

b Find the exact length of each side of the quadrilateral.

c Find its perimeter, correct to one decimal place.

d Calculate its area.

PATH
EXAMPLE 2
EXAMPLE 3

11.01

Technology

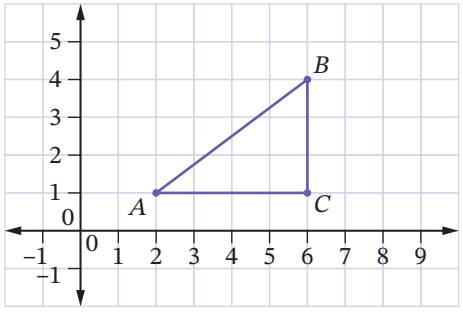
The length of an interval

In this activity, you will use dynamic geometry software to apply Pythagoras' theorem to find the length of an interval.

1 Insert a grid on the number plane, with x-values: 0 to 14 and y-values: 0 to 8.

2 Plot the points $A(2, 1)$, $B(6, 4)$ and $C(6, 1)$ on the grid. Join AB, BC and AC.

3 Find the lengths of AC and BC.



Foundation Standard Complex

4 Calculate $(AC)^2 + (BC)^2$ and $\sqrt{(AC)^2 + (BC)^2}$.

5 Now find the length of AB.

6 Compare answer **5** with answer **4**. What do you notice?

7 Move the number plane until the axes are centred in the middle of the screen. [Approx: x: –10 to 10 and y: –10 to 10]

8 Plot each pair of points and calculate the distance between them by repeating all the steps above.

a $X(-4, 2)$ and $Y(4, -4)$ **b** $P(-7, 5)$ and $Q(-2, -7)$

Did you know?

René Descartes



Alamy Stock Photo/IanDagnall Computing

The number plane is also called the Cartesian plane, named after its inventor, the French philosopher and mathematician René Descartes (1596–1650). In an age when Shakespeare and Galileo had made contributions to drama and science respectively, Descartes showed that many relationships and formulas could be represented by simple graphs. In 1637, he made a great contribution to mathematics when he published a book linking algebra and geometry for the first time.

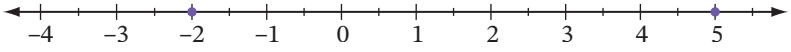
As a philosopher, the big question that Descartes spent many years thinking about was whether he, or the world, really existed. He finally concluded that he did exist, using a brilliant argument summarised by his famous quote.

Complete his famous quote: "I _____, therefore _____"

Investigation

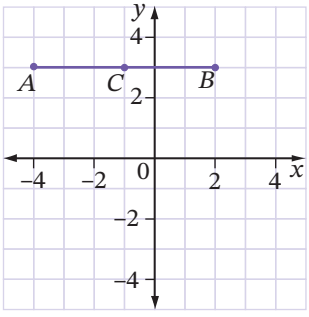
The midpoint of an interval

1 **a** What number is halfway between –2 and 5 on this number line?



b What is the average of –2 and 5?

2 **a** What are the coordinates of A and B on this number plane?



b What is the average of the x-values of A and B?

c What is the average of the y-values of A and B?

d What are the coordinates of C, the midpoint of AB?

e What do you notice about your answers for parts **b, c** and **d**?

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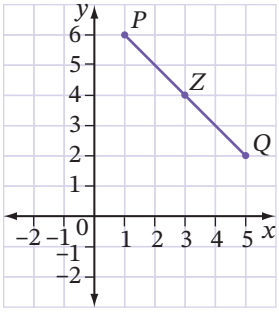
3 **a** What are the coordinates of P and Q?

b What is the average of the x-values of P and Q?

c What is the average of the y-values of P and Q?

d What are the coordinates of Z, the midpoint of PQ?

e How do the coordinates of Z relate to your answers for parts **b** and **c**?



4 Describe how to find the midpoint of any interval on a number plane given the coordinates of its endpoints.

Technology

The midpoint of an interval

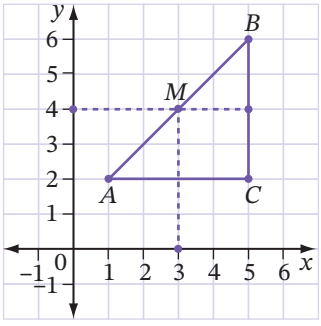
1 Use dynamic geometry software to create a number plane with x-values: 0 to 14 and y-values: 0 to 8.

2 Plot the points $A(1, 2)$, $B(5, 6)$ and $C(5, 2)$ and join AB, BC and AC.

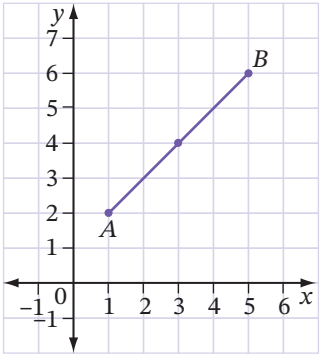
3 What is the average x-value on AC?

4 What is the average y-value on BC?

5 Label the midpoint of AB, M (as shown below). What are its coordinates?



6 Redraw interval AB. Select the midpoint and label it M. Does it have the same coordinates as you found in step **5**?



11.02 The midpoint of an interval

The **midpoint of an interval** is the point in the middle of an interval or halfway between 2 given points.

Technology
Finding midpoints

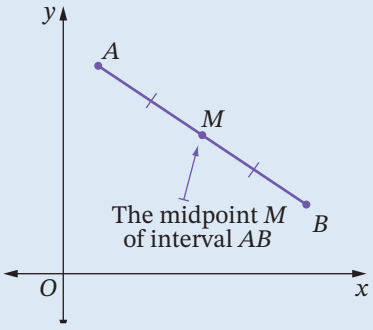
Spreadsheet
Finding midpoints

The midpoint of an interval

To find the midpoint of an interval joining 2 points A and B:

- calculate the average of the x-coordinates of A and B
- calculate the average of the y-coordinates of A and B

These 2 values give the coordinates of the midpoint.



Example 4

Find the midpoint of the interval joining:

a $A(5, -2)$ and $B(1, 6)$

b $P(-2, 7)$ and $Q(4, 0)$

Solution

a The x-coordinates are 5 and 1.

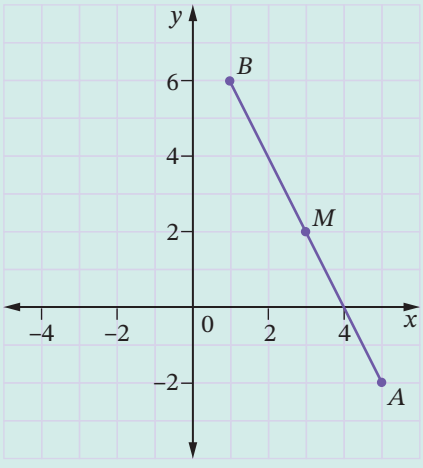
The average of 5 and 1 $= \frac{5+1}{2} = 3$.

The y-coordinates are -2 and 6.

The average of -2 and 6 $= \frac{-2+6}{2} = 2$

$\therefore$ The midpoint is the point $(3, 2)$.

The diagram shows that this is true.



b The x-coordinates are -2 and 4.

The average of -2 and 4 $= \frac{-2+4}{2} = 1$

The y-coordinates are 7 and 0.

The average of 7 and 0 $= \frac{7+0}{2} = 3\frac{1}{2}$

$\therefore$ The midpoint is the point $\left(1, 3\frac{1}{2}\right)$.

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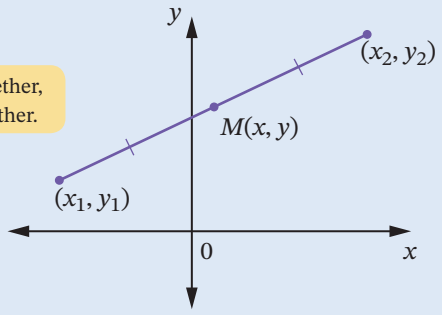
The midpoint formula

The midpoint, M, of the interval joining 2 points (x_1, y_1) and (x_2, y_2) has coordinates:

$$M(x, y) \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Add the x-coordinates together, add the y-coordinates together.

We've used '$\equiv$' ('is identical to') rather than '$=$' because we are referring to points (x, y), not numbers.



Example 5

Use the midpoint formula to find the midpoint of the interval joining $P(12, -8)$ and $Q(7, -2)$.

Solution

For $P(12, -8)$ and $Q(7, -2)$:

$x_1 = 12, y_1 = -8, x_2 = 7, y_2 = -2$

Midpoint, $M \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Add the x-coordinates together, add the y-coordinates together.

$$\equiv \left(\frac{12+7}{2}, \frac{-8+(-2)}{2} \right)$$

$$\equiv \left(\frac{19}{2}, \frac{-10}{2} \right)$$

$$\equiv \left(9\frac{1}{2}, -5 \right)$$

PATH

Video Distance, midpoint and gradient formulas

11.02

Technology Midpoint and distance between 2 points

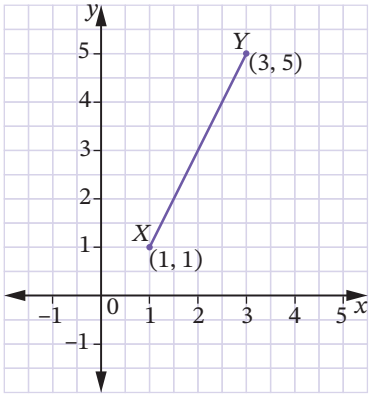
Spreadsheet Midpoint and distance between 2 points

EXERCISE 11.02 ANSWERS ON P. 644

The midpoint of an interval U F R C

1 What is the midpoint of interval XY?
Select the correct answer **A**, **B**, **C** or **D**.

A (2, 3) **B** (2, 4) **C** (3, 2) **D** (3, 3)



Foundation Standard Complex

2 On a number plane, plot and join each pair of points, then find and plot the midpoint of each interval.

a (1, 3) and (7, 5) **b** (0, 2) and (10, 4) **c** (−3, 3) and (5, −1)

EXAMPLE 4, 5

3 Find the midpoint of the interval joining:

a (10, 10) and (6, 4) **b** (2, 7) and (12, 9) **c** (0, 5) and (9, 3)

d (2, 2) and (2, 4) **e** (−2, 6) and (10, 3) **f** (9, −1) and (5, 2)

g (−8, 11) and (9, −4) **h** (−6, −1) and (0, 0) **i** (3, −3) and (−5, −10)

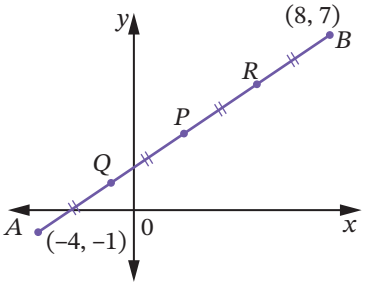
j (6, −1) and (−8, −1) **k** (5, −6) and (9, −10) **l** (−8, −7) and (−3, −5)

4 The interval AB has been subdivided into 4 equal parts. Find the coordinates of:

a P

b Q

c R



5 What is the midpoint of $K(6, 9)$ and $L(-2, 7)$? Select **A**, **B**, **C** or **D**.

A (4, 16) **B** (−4, 2) **C** (−2, 1) **D** (2, 8)

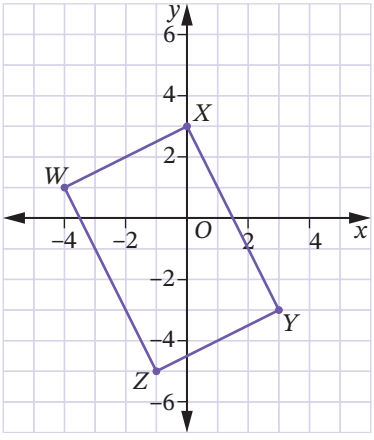
6 $WXYZ$ is a rectangle.

a Find the midpoint of:

i diagonal WY

ii diagonal XZ.

b What do you notice about the 2 midpoints in part **a**? Which property for the diagonals of a rectangle does this show? R C



7 The diagram shows an isosceles triangle, PQR.

a Find M, the midpoint of PQ.

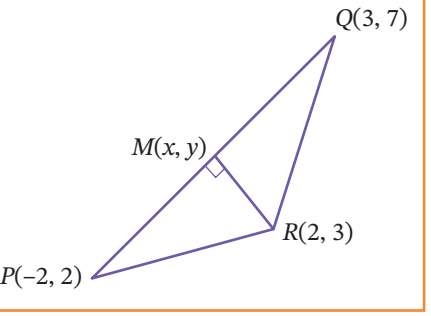
b Find, as a surd, the length of:

i PQ **ii** MR **iii** QR

c Calculate, correct to one decimal place

i the area of $\triangle PQR$

ii the perimeter of $\triangle PQR$.



Foundation Standard Complex

9780170479585

8 The vertices of a square are $K(1, 6)$, $L(7, 2)$, $M(3, -4)$ and $P(-3, 0)$.

- **a** Find the length of the diagonal KM.
- **b** Find the length of the diagonal LP.
- **c** Are the diagonals equal in length?
- **d** Find the coordinates of the midpoint of KM.
- **e** Find the coordinates of the midpoint of LP.
- **f** Do the diagonals bisect each other? Explain.

The gradient of a line

11.03

The **gradient** of a line (or interval) is a measure of how steep the line is. It is represented by the variable m and is the ratio (fraction) of the vertical **rise** to the horizontal **run** between any 2 points on a line.

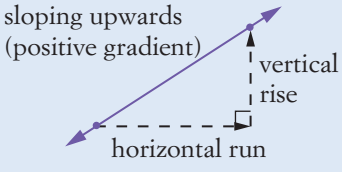
Worksheets
Drawing gradients
A page of intervals
A page of lines
Gradient between two points

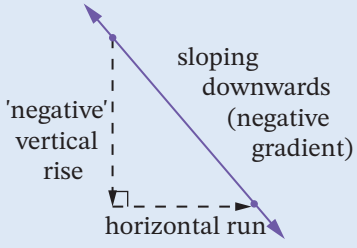
Puzzle
Find the gradient 1

The gradient of a line or interval

$$m = \frac{\text{vertical rise}}{\text{horizontal run}} = \frac{\text{rise}}{\text{run}}$$

- A line **sloping upwards** has a **positive rise** and a **positive gradient**.
- A line **sloping downwards** has a **negative rise** and a **negative gradient**.
- The **run** is always **positive**.

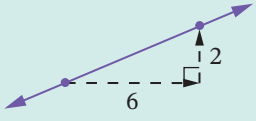




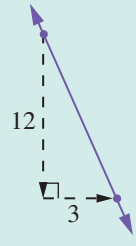
Example 6

Calculate the gradient of each line.

a



b



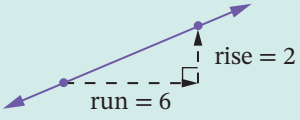
Foundation | Standard | Complex

Solution

a The line has a positive rise (↑).

$m = \frac{2}{6}$ $\quad m = \frac{\text{rise}}{\text{run}}$

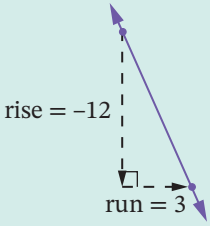
$= \frac{1}{3}$



b The line has a negative rise or a drop (↓).

$m = \frac{-12}{3}$ $\quad m = \frac{\text{rise}}{\text{run}}$

$= -4$



Videos
The gradient of a line

Gradient (slope) of a line

Example 7

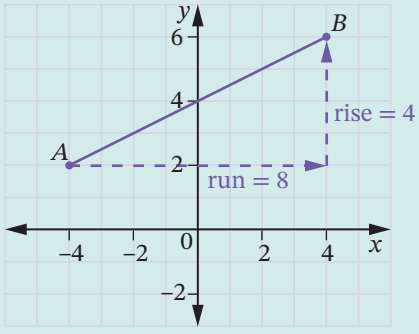
Calculate the gradient of the line going through:

a $A(-4, 2)$ and $B(4, 6)$ **b** $C(-2, 4)$ and $D(2, -4)$.

Solution

Form a right-angled triangle with the interval as the hypotenuse, as shown.

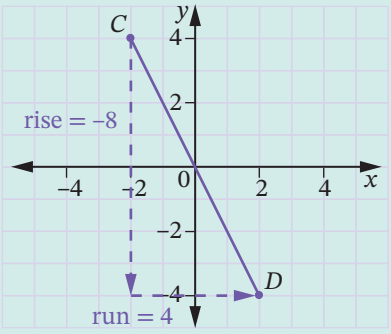
a



$m = \frac{4}{8}$

$= \frac{1}{2}$

b



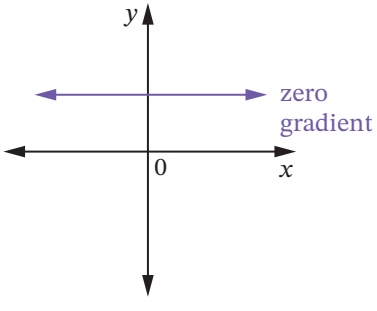
$m = \frac{-8}{4}$

$= -2$

The size and sign of the gradient

The **size** of the gradient shows how quickly the line is going up or down: the higher the number, the steeper the line. The **sign** of the gradient shows whether the line is sloping upwards (positive) or downwards (negative) as you look from left to right.

A **horizontal (flat) line** has **zero gradient**.



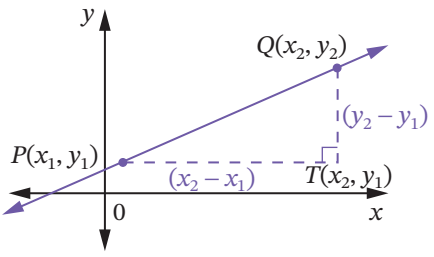
The gradient formula

Consider the line passing through the points $P(x_1, y_1)$ and $Q(x_2, y_2)$.

Vertical rise = difference in y-coordinates $= y_2 - y_1$

Horizontal run = difference in x-coordinates $= x_2 - x_1$

Gradient, $m = \dfrac{\text{difference in } y}{\text{difference in } x} = \dfrac{y_2 - y_1}{x_2 - x_1}$



The gradient formula

The gradient (m) of a line or interval joining 2 points (x_1, y_1) and (x_2, y_2) is given by the formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Subtract the y-coordinates **on top**, subtract the x-coordinates **below**.

Example 8

Use the gradient formula to calculate the gradient of the line going through $A(-4, 2)$ and $B(4, 6)$ (from Example **7 a**).

Solution

For $A(-4, 2)$ and $B(4, 6)$:

$x_1 = -4, y_1 = 2, x_2 = 4, y_2 = 6$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$
$$= \frac{6 - 2}{4 - (-4)}$$
$$= \frac{4}{8}$$
$$= \frac{1}{2}$$

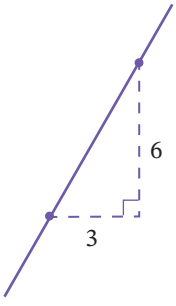
Subtract the y-coordinates **on top**, subtract the x-coordinates **below**.

EXERCISE 11.03 ANSWERS ON P. 644

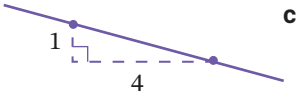
The gradient of a line UFRC

1 State whether the gradient of each line is positive (+) or negative (−).

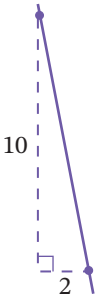
a



b



c



PATH

Video Distance, midpoint and gradient formulas

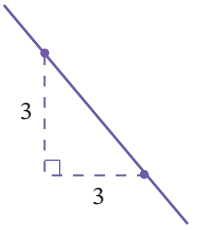
Worksheet Gradient, midpoint and distance

Puzzle Intervals match-up

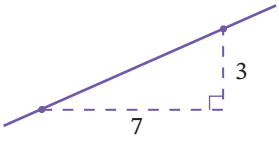
11.03

Foundation | Standard | Complex

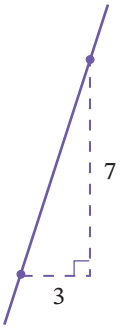
d



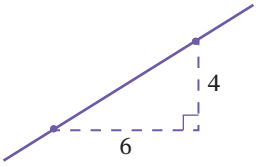
e



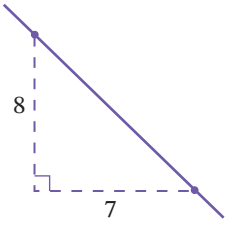
f



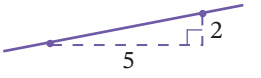
g



h



i

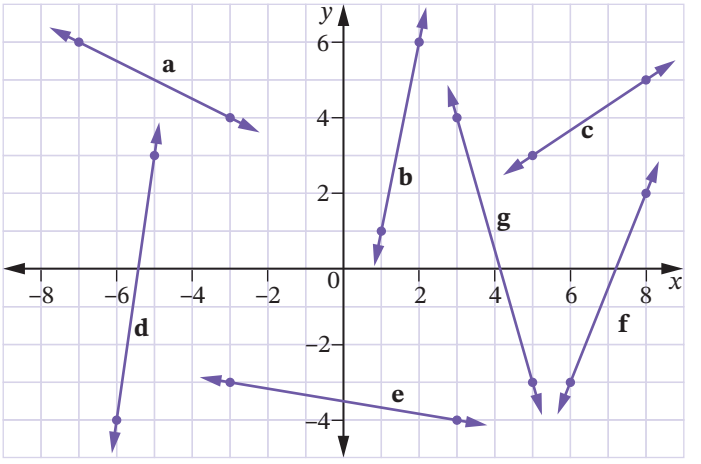


EXAMPLE 6

2 Calculate the gradient of each line in question **1**.

EXAMPLE 7

3 Calculate the gradient of each interval.



4 Which pair of points form an interval that has a gradient of 2?
Select the correct answer **A**, **B**, **C** or **D**.

A (0, 2), (2, 0) **B** (−5, 2), (5, 7) **C** (−2, −2), (1, 4) **D** (−6, −8), (−2, −6)

EXAMPLE 8

5 Calculate the gradient of the interval joining each pair of points, either by plotting the points on a number plane or by using the gradient formula.

a $A(1, 4)$ and $B(3, 6)$ **b** $Z(4, 7)$ and $T(1, 6)$ **c** $M(4, -2)$ and $D(3, 3)$

d $A(-2, -1)$ and $P(4, -4)$ **e** $D(0, -1)$ and $F(3, -1)$ **f** $H(9, -2)$ and $L(-1, 3)$

g $K(4, 9)$ and $M(-4, -5)$ **h** $Z(-8, -6)$ and $T(4, 0)$ **i** $F(0, 4)$ and $B(3, 4)$

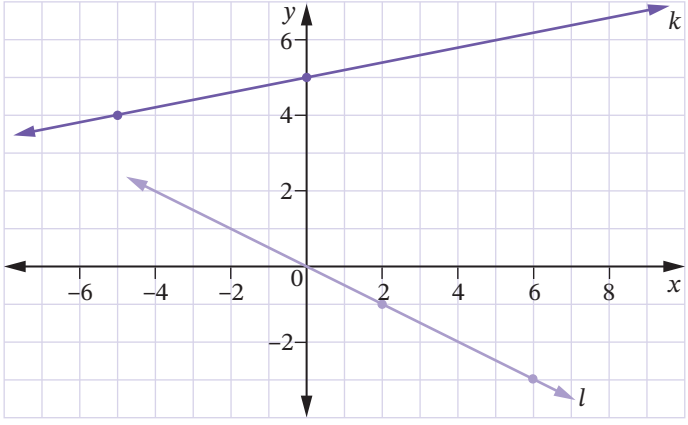
Foundation Standard Complex

9780170479585

6 Which pair of points form an interval that has a gradient of –1? Select **A**, **B**, **C** or **D**.

A (1, 2), (2, 1)
B (–5, 6), (6, 7)
C (2, 2), (3, –1)
D (–5, –10), (4, –11)

7 Find the gradient of the lines labelled k and l.



8 Which pair of points lie on a line with a gradient of 0? Select **A**, **B**, **C** or **D**.

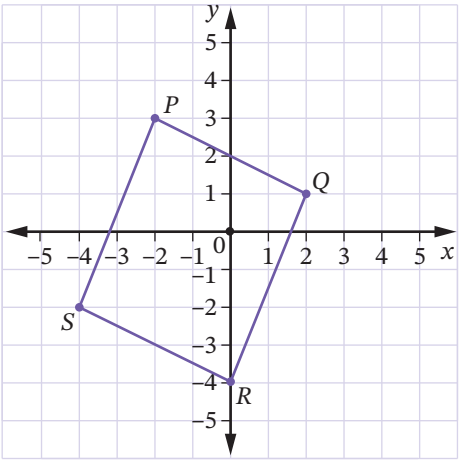
A (3, 3), (–3, –3)
B (–5, 7), (5, 7)
C (1, –6), (1, 4)
D (0, 9), (9, 0)

9 $PQRS$ is a quadrilateral. R C

a Find the gradient of each side of $PQRS$.

b Which sides have the same gradient? What does that mean about those sides?

c What type of quadrilateral is $PQRS$?



10 A triangle is formed by joining $P(-8, -2)$, $Q(-2, 10)$ and $R(4, 6)$. T is the midpoint of PQ, and Z is the midpoint of QR. R C

a Draw a diagram showing all the given information.

b Find the coordinates of T and Z.

c Calculate the gradient of interval TZ and side PR. What do you notice?

d Calculate the length of interval TZ and side PR. What do you notice?

Did you know?

Stairs, ramps and hills

- Some stairs are easier to climb than others: the gradient of stairs should not be more than 5 in 6 $\left(\frac{5}{6}\right)$.
- A ramp for walking should not have a gradient higher than $\frac{3}{10}$ and the gradient of a wheelchair ramp should be between $\frac{1}{14}$ and $\frac{1}{8}$.
- The maximum slope of a hill for parking your car safely is $\frac{2}{9}$.
- The steepest street in the world is Baldwin Street in Dunedin, New Zealand, with one section having a gradient of 35% or $\frac{7}{20}$.

Would it be safe to park your car on Bulli Pass in Wollongong, NSW, which has a slope of $\frac{1}{6}$?



Shutterstock.com/Deyan Denchev

Baldwin St, the steepest street in the world, in Dunedin, New Zealand.

Technology

The gradient of collinear points

In this activity, you will use dynamic geometry software to investigate collinear points.

1 Create a number plane with x-values: 0 to 30 and y-values: 0 to 16.

2 Plot the points $A(5, 2)$, $B(9, 5)$ and $C(17, 11)$. Create intervals AB and BC.

3 To check if the 3 points are collinear, select **Slope** for interval AB, and again for interval BC. Is the slope (gradient) the same for AB and BC? If so, these points are called **collinear points**. What do you think **collinear points** are? Discuss this with other students and explain it in your own words.

4 Now test whether each set of points are collinear.

a (0, 3), (2, 4) and (6, 6)

b (8, 0), (−8, 0) and (−6, −10)

c (1, 8), (2, 9) and (4, 10)

d (−2, 8), (−1, 6) and (0, 1)

e (2, 7), (4, 10), (5, 13) and (6, 17)

f (−14, 11), (−8, 9), (1, 6) and (16, 1)

Graphing linear equations

11.04

A relationship between 2 variables such as x and y whose graph is a **straight line** is called a **linear relationship**. The expression of that relationship as an algebraic formula, such as $y = x + 2$, is called a **linear equation**.

Example 9

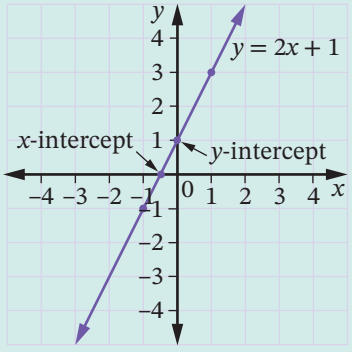
Graph $y = 2x + 1$ on a number plane.

Solution

Complete a table of values. Choose x-values close to 0 for easy calculation and graphing.

x	−1	0	1
y	−1	1	3

Graph the table of values on a number plane. Rule a straight line through the points, place arrows at each end, and label the line with its equation.



Note:

- the ***x*-intercept** of the line is $-\frac{1}{2}$: this is the x-value where the line cuts the ***x*-axis**
- the ***y*-intercept** of the line is 1: this is the y-value where the line cuts the ***y*-axis**
- every point on the line follows the linear equation $y = 2x + 1$: for example, (−4, −7), (−2, −3) and (2, 5) lie on the line and also follow the rule $y = 2x + 1$
- there are an infinite number of points that follow the rule: arrows on both ends of the line indicate that it has infinite length.

Using *x*- and *y*-intercepts to graph lines

We can also graph a linear equation by finding its x- and y-intercepts first.

Since any point on the x-axis has a y-coordinate of 0, we can substitute $y = 0$ into the equation to find the x-intercept.

Similarly, any point on the y-axis has an x-coordinate of 0, so we can substitute $x = 0$ into the equation to find the y-intercept.

Using *x*- and *y*-intercepts to graph lines

- To find the x-intercept, substitute $y = 0$ and solve the equation.
- To find the y-intercept, substitute $x = 0$ and solve the equation.

Video
Graphing linear equations

Worksheets
Graphing linear equations
A page of intervals
A page of lines
A page of number planes
Number plane grid paper

Interactives
Graphing lines
Function builder: Basics
Function builder

Example 10

Find the x- and y-intercepts for $y = -4x + 8$ to draw its graph.

Solution

For x-intercept, $y = 0$:

$0 = -4x + 8$

$4x = 8$

$x = 2$

The x-intercept is 2.

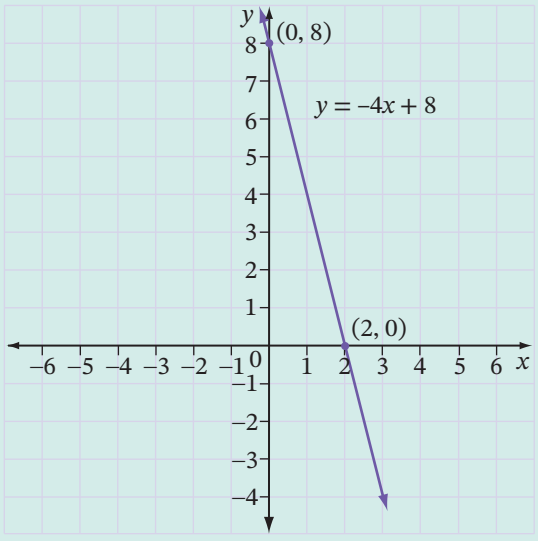
For y-intercept, $x = 0$:

$y = -4 \times 0 + 8$

$= 8$

The y-intercept is 8.

Plot both intercepts on the axes, draw a line through the 2 points and label the line with its equation.



Worksheets
Straight-line equations

Testing if a point lies on a line

A point lies on a line if its (x, y) coordinates satisfy the equation of the line.

- Separate the equation into LHS (left-hand side) and RHS (right-hand side)
- Substitute the coordinates of the point into both sides
- If LHS = RHS, the point satisfies the equation and lies on the line
- If LHS ≠ RHS, the point does not lie on the line.

Video
Testing if a point lies on a line

Example 11

Test whether each point lies on the line $y = 2x + 3$.

a (3, 9) **b** (8, –4)

Solution

a Substitute $x = 3$ and $y = 9$ into $y = 2x + 3$.

LHS $= y$ RHS $= 2x + 3$

$= 9$ $= 2 \times 3 + 3$

$= 9$

LHS = RHS, so (3, 9) lies on the line.

b Substitute $x = 8$ and $y = -4$ into $y = 2x + 3$.

LHS $= y$ RHS $= 2x + 3$

$= -4$ $= 2 \times 8 + 3$

$= 19$

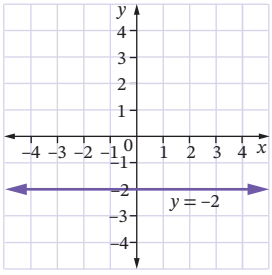
LHS ≠ RHS, so (8, –4) does not lie on the line.

Horizontal and vertical lines

A **horizontal line** is flat and runs across (sideways), parallel to the x-axis.

Every point on the horizontal line has y-coordinate -2, and the y-intercept is -2.

x	-1	0	1	2
y	-2	-2	-2	-2

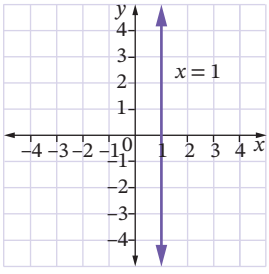


y is always equal to -2, so the equation of this horizontal line is $y = -2$.

A **vertical line** runs up and down, parallel to the y-axis and at right angles to a horizontal line.

Every point on the vertical line has x-coordinate 1, and the x-intercept is 1.

x	1	1	1	1
y	0	1	2	3



x is always equal to 1, so the equation of this vertical line is $x = 1$.

Horizontal and vertical lines

The **equation of a horizontal line** is of the form $\boldsymbol{y = c}$ (where c is a constant number).

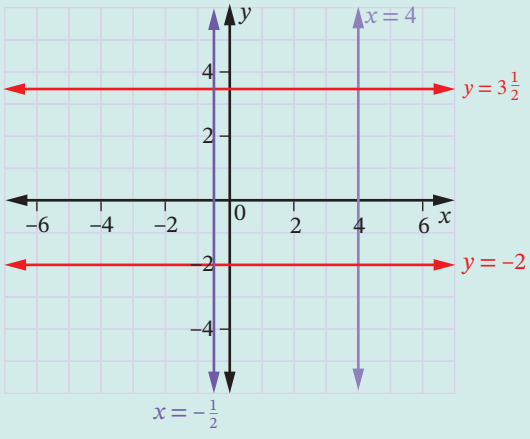
The **equation of a vertical line** is of the form $\boldsymbol{x = c}$ (where c is a constant number).

Example 12

Graph each line on the same number plane.

a $x = 4$ **b** $y = -2$ **c** $x = -\frac{1}{2}$ **d** $y = 3\frac{1}{2}$

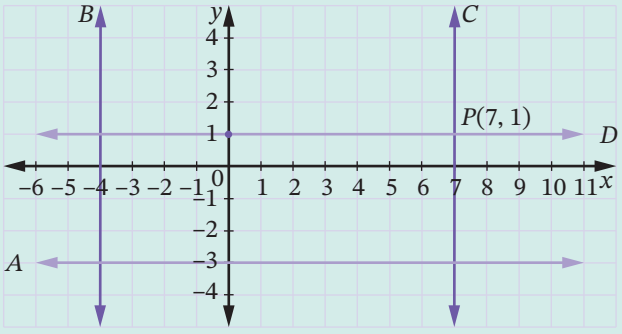
Solution



Video
Horizontal and vertical lines

Example 13

Find the equations of lines A, B, C and D.



Solution

- Line A is horizontal and has a y-intercept of -3, so its equation is $y = -3$.
- Line B is vertical and has an x-intercept of -4, so its equation is $x = -4$.
- Line C is vertical and goes through $(7, 1)$, so its equation is $x = 7$.
- Line D is horizontal and goes through $(7, 1)$, so its equation is $y = 1$.

EXERCISE 11.04 ANSWERS ON P. 645

Graphing linear equations UFRC

EXAMPLE 9

1 Graph each linear equation on a number plane, and write:

i its x-intercept **ii** its y-intercept.

a	$y = x - 3$	**b**	$y = 2x + 2$	**c**	$y = -x + 5$	**d**	$y = 3x$
e	$y = -x$	**f**	$y = 10 - 4x$	**g**	$y = \frac{x+2}{2}$	**h**	$y = 2x + 1$
i	$y = \frac{x}{3} - 2$	**j**	$y = \frac{3x}{4}$	**k**	$x + y = 4$	**l**	$x - y = 2$

2 What is the x-intercept of the graph of $y = 6 - 3x$? Select the correct answer **A**, **B**, **C** or **D**.

A -3 **B** 2 **C** 3 **D** 6

EXAMPLE 10

3 Graph each linear equation after finding its x- and y-intercepts.

a	$y = 3x - 6$	**b**	$y = \frac{6-x}{2}$	**c**	$x + y = 7$
d	$x - y = 6$	**e**	$y = \frac{2x}{3} + 1$	**f**	$3x - 4y = 12$
g	$x + y = 4$	**h**	$2x - y = 5$	**i**	$\frac{x}{2} + \frac{y}{3} = 1$

Foundation Standard Complex

9780170479585

EXAMPLE 11

4 Test whether the point (3, –1) lies on each line:

a $y = 2x - 7$ **b** $y = -2x + 4$ **c** $x - y = 4$

d $x + 2y = 1$ **e** $y = -3x + 8$ **f** $x - 3y = 0$

5 Which of these points lies on the line $x - y = -2$? Select **A**, **B**, **C** or **D**.

A (1, 1) **B** (–1, –1) **C** (0, 2) **D** (–2, 4)

EXAMPLE 12

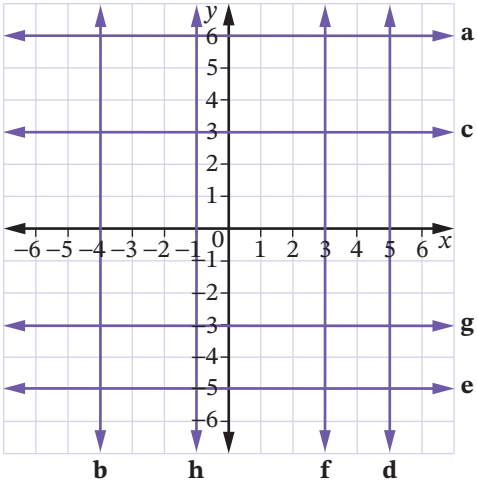
6 Graph each set of lines on a number plane.

a $x = 2, x = -3$ **b** $y = 6, y = -2$

c $x = 2\frac{1}{2}, x = 0, y = -5$ **d** $y = 1.5, y = 3, x = -2$

EXAMPLE 13

7 Write down the equation of each line shown.



8 **a** Show that the point (3, 2) lies on both the lines $x = 2y - 1$ and $x + y = 5$. R C

b So at what point do the lines $x = 2y - 1$ and $x + y = 5$ cross?

9 Show that the lines $y = -3x + 6$ and $y = -2x$ intersect at the point (6, –12). R C

10 Find the equation of each line described. R C

a A horizontal line that passes through the y-axis at 6.

b A vertical line that goes through (–5, 2).

c Each point on the line is 2 units above the x-axis.

d Each point on the line is 5 units to the left of the y-axis.

e The line is parallel to the x-axis and passes through the point (0, –2).

f A horizontal line that goes through (4, –1).

g The line goes through the points (–1, 8) and (–1, 2).

11 What is the equation of: R C

a the y-axis? **b** the x-axis?

Foundation Standard Complex

Technology

Graphing linear equations

In this activity, we will use dynamic geometry software to graph a linear equation.

1 Create a number plane with axes: x: –10 to 10 and y: –10 to 10.

2 Enter the following linear equations in column A of the spreadsheet.

A1: $y = 3x + 1$ A2: $y = 3x$ A3: $y = 2x + 4$

A4: $y = -2x - 5$ A5: $y = x + 2$ A6: $y = -x - 1$

3 Read the gradient of each line by selecting the **Slope tool**. Select a line and you will see the right-angled triangle showing $\frac{\text{rise}}{\text{run}}$.

4 Enter each linear equation and find the gradient of each line.

a $y = x + 1$ **b** $y = -2x - 4$ **c** $y = 6x + 3$

d $y = 3x$ **e** $y = -8x$ **f** $y = -4x - 6$

Quiz
Mental skills 11

Mental skills 11: Maths without calculators ANSWERS ON P. 647

Divisibility tests

How can you tell if a number is divisible by 2? Look at its last digit. If that digit is 2, 4, 6, 8 or 0, then the number is divisible by 2 (that is, it is even).

How can you tell if a number is divisible by 5? If its last digit is 0 or 5, then the number is divisible by 5.

These are examples of **divisibility tests**—rules for checking whether a number is divisible by a certain number. The table below shows some common divisibility tests.

A number is divisible by:	if:
2	its last digit is 2, 4, 6, 8 or 0
3	the sum of its digits is divisible by 3
4	its last 2 digits form a number divisible by 4
5	its last digit is 0 or 5
6	it is even and the sum of its digits is divisible by 3
9	the sum of its digits is divisible by 9
10	its last digit is 0

1 Study each example.

a Test whether 748 is divisible by 2, 3 or 4.

- Last digit is 8 (even). ∴ 748 is divisible by 2.
- Sum of digits = 7 + 4 + 8 = 19, which is not divisible by 3. ∴ 748 is not divisible by 3.
- 48 is divisible by 4. ∴ 748 is divisible by 4 (748 ÷ 4 = 187).

b Test whether 261 is divisible by 5 or 9.

- Last digit is 1, not 0 or 5. ∴ 261 is not divisible by 5.
- 2 + 6 + 1 = 9, which is divisible by 9. ∴ 261 is divisible by 9 (261 ÷ 9 = 29).

c Test whether 570 is divisible by 4, 6 or 10.

- 70 is not divisible by 4. ∴ 570 is not divisible by 4.
- 570 is even and 5 + 7 + 0 = 12, which is divisible by 3. ∴ 570 is divisible by 6 (570 ÷ 6 = 95).
- Last digit is 0. ∴ 570 is divisible by 10 (570 ÷ 10 = 57).

2 Test whether each number is divisible by 2, 3, 5 or 6.

a	250	**b**	189	**c**	78	**d**	465
e	1024	**f**	840	**g**	715	**h**	627

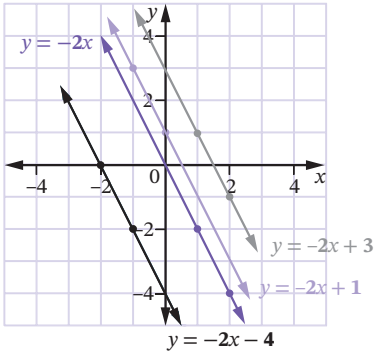
3 Test whether each number is divisible by 4, 9 or 10.

a	144	**b**	280	**c**	522	**d**	4170
e	936	**f**	726	**g**	342	**h**	5580

Investigation

Comparing gradients and *y*-intercepts

1 This number plane shows the graphs of 4 linear equations.



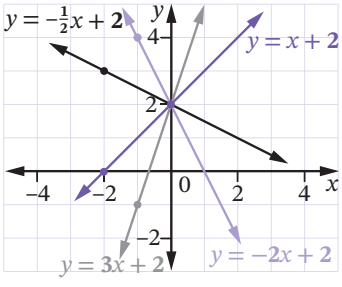
a Copy and complete this table by finding the gradient and *y*-intercept of each line.

Equation	Gradient	*y*-intercept
$y = -2x - 4$	−2	−4
$y = -2x$		
$y = -2x + 1$		
$y = -2x + 3$		

b How are the 4 lines:

i similar? **ii** different?

2 This number plane shows the graphs of another 4 linear equations.



a Copy and complete this table.

Equation	Gradient	*y*-intercept
$y = -2x + 2$		
$y = -\frac{1}{2}x + 2$		
$y = x + 2$		
$y = 3x + 2$		

b How are the 4 lines:

i similar? **ii** different?

3 Copy and complete this table for the 3 lines graphed.

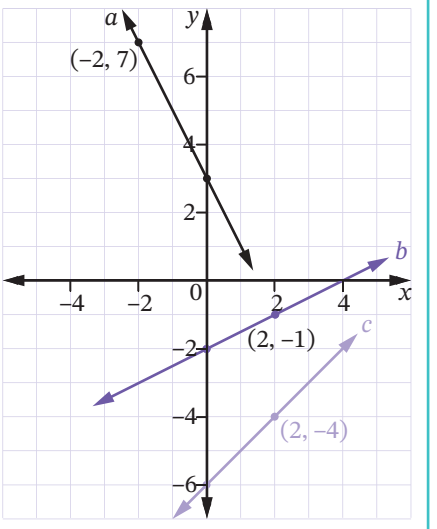
a

Equation	Gradient	*y*-intercept
$y = -2x + 3$		
$y = \frac{1}{2}x - 2$		
$y = x - 6$		

b Explain how the values for the gradient and *y*-intercept of each line relate to its equation.

c What is the equation of a line that has a gradient of 4 and a *y*-intercept of –1?

d What is the gradient and *y*-intercept of the line with equation $y = -x + 7$?



11.05 The gradient–intercept equation $y = mx + c$

Puzzle
Equation of a line

Interactive
Graphing slope-intercept

Spreadsheet
Drawing straight lines: $y = mx + c$

The gradient–intercept form of a linear equation

The equation of a straight line is $y = mx + c$, where m is the gradient and c is the *y*-intercept.

For this reason, $y = mx + c$ is also called the **gradient–intercept form** of a linear equation.

Example 14

Find the gradient and *y*-intercept of the line with equation:

a $y = 4x - 7$ **b** $y = \frac{1}{2}x + 3$ **c** $y = 3 - 5x$ **d** $y = \frac{3x}{4} - 1$

Solution

a $y = 4x - 7$ is in the form $y = mx + c$ $\therefore$ Gradient $m = 4$ and *y*-intercept $c = -7$

b For $y = \frac{1}{2}x + 3$, gradient $m = \frac{1}{2}$ and *y*-intercept $c = 3$

c $y = 3 - 5x$ can be rewritten as $y = -5x + 3$ $\therefore$ Gradient $m = -5$ and *y*-intercept $c = 3$

d For $y = \frac{3x}{4} - 1$, gradient $m = \frac{3}{4}$ and *y*-intercept $c = -1$

Example 15

Write the equation of the line that has a gradient of –4 and a y-intercept of –6.

Solution

$m = -4, c = -6$

$\therefore$ The equation of the line is $y = -4x - 6$. Using $y = mx + c$.

Example 16

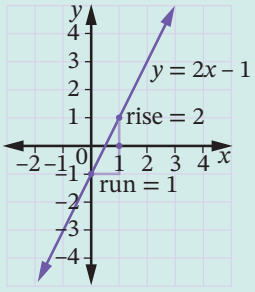
Graph each linear equation by finding the gradient and y-intercept first.

a $y = 2x - 1$ **b** $y = -2x + 5$ **c** $y = \frac{3}{4}x - 2$

Solution

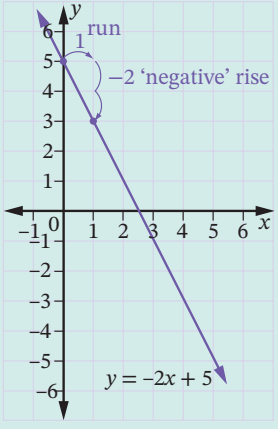
a $y = 2x - 1$ has a gradient of 2 and a y-intercept of –1.

- Plot the y-intercept –1 on the y-axis.
- Make a gradient of 2 by moving **across** 1 unit (**run**) and **up** 2 units (**rise**) and marking the point at (1, 1).
- Rule a line through this point and the y-intercept.
- Label the line with the equation $y = 2x - 1$



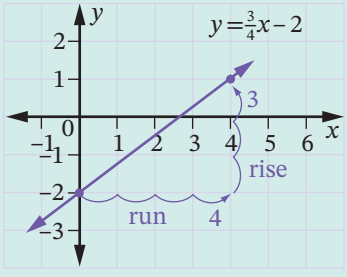
b $y = -2x + 5$ has a gradient of –2 and a y-intercept of 5.

- Plot the y-intercept 5 on the y-axis.
- Make a gradient of –2 by moving **across** 1 unit (**run**) and **down** 2 units (‘**negative**’ **rise**) and marking the point at (1, 3).
- Rule a line through this point and the y-intercept.



c $y = \frac{3}{4}x - 2$ has a gradient of $\frac{3}{4}$ and a y-intercept of –2.

- Plot the y-intercept –2 on the y-axis.
- Make a gradient of $\frac{3}{4}$ by moving **across** 4 units (**run**) and **up** 3 units (**rise**) and marking the point at (4, 1).
- Rule a line through this point and the y-intercept.



11.05

Video
Graphing
$y = mx + c$

EXERCISE 11.05 ANSWERS ON P. 647

The gradient–intercept equation $y = mx + c$ U F

EXAMPLE 14

1 Find the gradient and y-intercept of the line with equation:

a $y = 3x - 2$ **b** $y = 2x + 7$ **c** $y = x + 4$ **d** $y = -x - 9$

e $y = \frac{3x}{4} + 6$ **f** $y = x$ **g** $y = \frac{x}{2} - 11$ **h** $y = \frac{2}{3}x + 6$

i $y = 8 - 2x$ **j** $y = 5 - x$ **k** $y = \frac{x}{4}$ **l** $y = -3x + 11$

m $y = 5$ **n** $y = \frac{2}{5}x + 1$ **o** $y = x - \frac{7}{2}$ **p** $y = \frac{3x}{4} + 1$

2 Which equation has a gradient of 5? Select the correct answer **A**, **B**, **C** or **D**.

A $y = x + 5$ **B** $y = 5 - x$ **C** $y = 5$ **D** $y = 5x$

EXAMPLE 15

3 Write the equation of the line with:

a a gradient of 2 and a y-intercept of -1 **b** a gradient of $\frac{3}{4}$ and a y-intercept of 2

c a gradient of -3 and a y-intercept of 5 **d** a gradient of $-\frac{2}{5}$ and a y-intercept of 3

4 For each pair of lines, state which line is steeper.

a $y = 2x + 3$ or $y = x + 7$ **b** $y = -3x - 5$ or $y = -2x + 1$

c $y = -\frac{1}{2}x + 3$ or $y = -x + 3$ **d** $y = -5x - 2$ or $y = 4x + 2$

e $y = -2x + 7$ or $y = 2x - 6$ **f** $y = -4x - 5$ or $y = \frac{7}{3}x + 1$

g $y = 8 + \frac{5}{4}x$ or $y = x + 2$ **h** $y = 2x$ or $y = 6 - 2x$

EXAMPLE 16

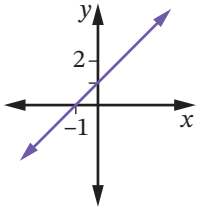
5 Graph each linear equation by finding the gradient and y-intercept first.

a $y = 2x + 1$ **b** $y = x - 4$ **c** $y = 3x + 2$ **d** $y = -4x$

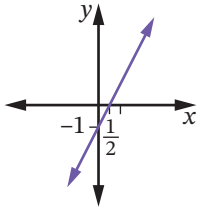
e $y = -3x - 3$ **f** $y = \frac{x}{2} + 2$ **g** $y = 8 - \frac{3x}{4}$ **h** $y = \frac{5x}{2} - 1$

6 Which diagram shows the graph of $y = 2x - 1$? Select **A**, **B**, **C** or **D**.

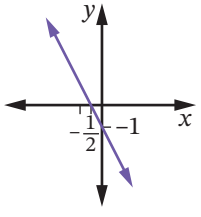
A



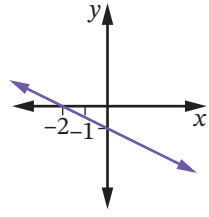
B



C



D



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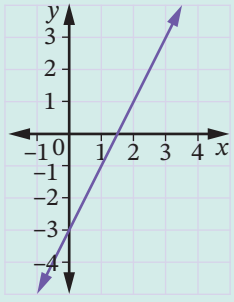
Finding the equation of a line

11.06

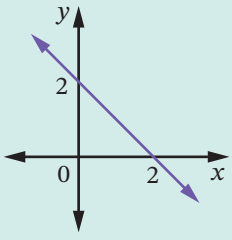
Example 17

Find the equation of each line.

a

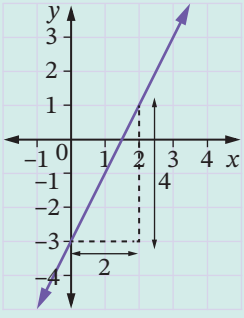


b



Solution

a Select 2 points on the line to find the gradient, say (0, −3) and (2, 1).



Gradient $m = \frac{\text{rise}}{\text{run}} = \frac{4}{2} = 2$

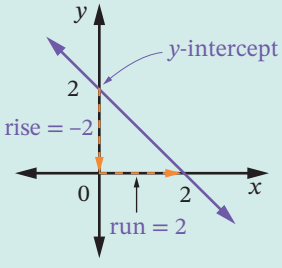
y-intercept $c = -3$ from graph

$\therefore$ The equation of the line is $y = 2x - 3$ $y = mx + c$

We can check that this equation is correct for another point, say (3, 3).

When $x = 3$, $y = 2 \times 3 - 3 = 3$

b Find the gradient of the line passing through (0, 2) and (2, 0).



Gradient $m = \frac{\text{rise}}{\text{run}} = \frac{-2}{2} = -1$

y-intercept $c = 2$ from graph

$\therefore$ The equation of the line is $y = -x + 2$ $y = mx + c$

Video
Finding the equation of a line

Worksheets
Finding the equation of a line
A page of intervals
A page of lines

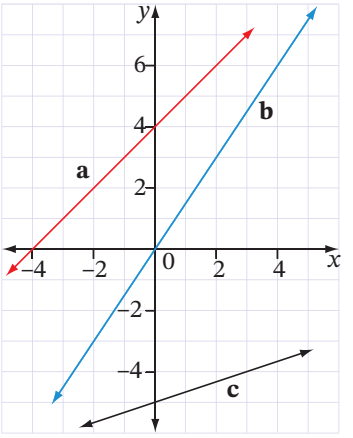
Puzzle
Straight-line equations

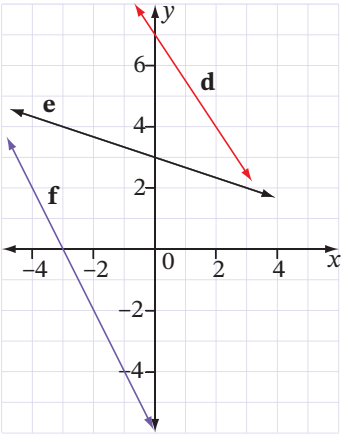
EXERCISE 11.06 ANSWERS ON P. 648

Finding the equation of a line U F

EXAMPLE 17

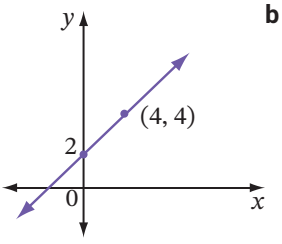
1 Find the equation of each line.



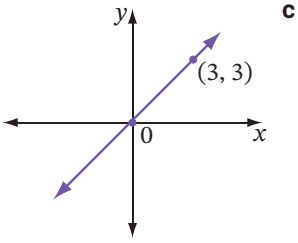


2 Find the equation of each line.

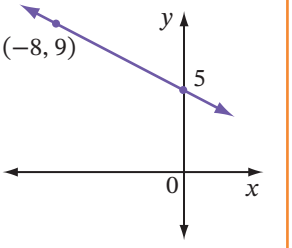
a



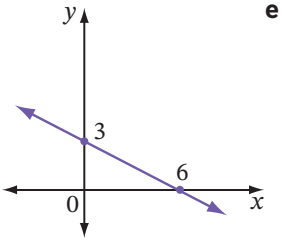
b



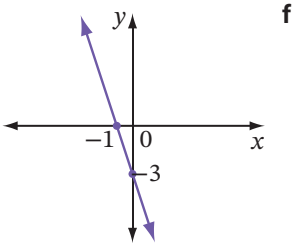
c



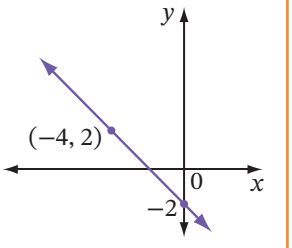
d



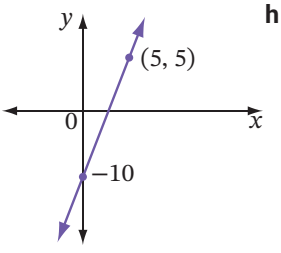
e



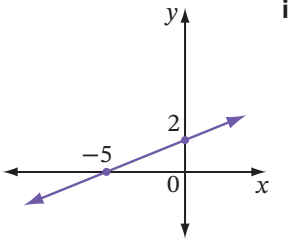
f



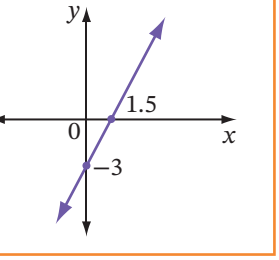
g



h



i



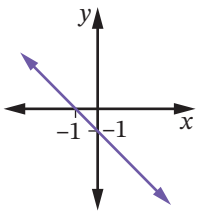
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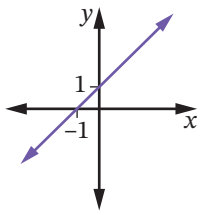
3 Match each linear equation to its graph.

a $y = x - 1$ **b** $y = x + 1$ **c** $y = -x - 1$ **d** $y = -x + 1$

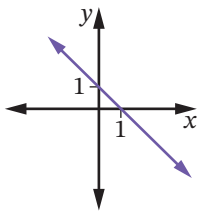
A



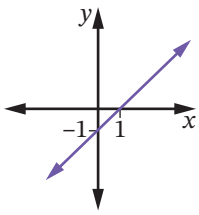
B



C



D



11.07

Solving linear equations graphically 11.07

In Chapter 7, *Equations*, we solved equations **algebraically**. We can also solve an equation **graphically** by first graphing it on the number plane.

Example 18

Solve the equation $2x - 3 = 5$ graphically.

Solution

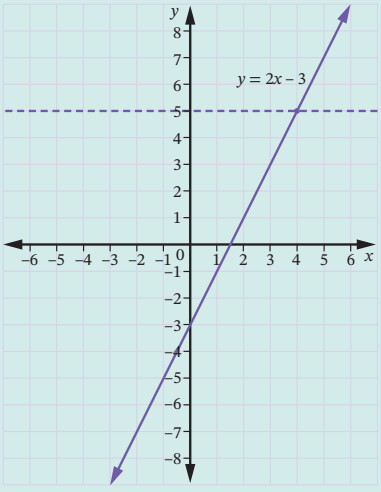
Graph $y = 2x - 3$ (the LHS of the equation) on a number plane.

For every point on the line, the y-coordinate is the value of $2x - 3$. So to solve $2x - 3 = 5$, we need to find the point whose y-coordinate is 5.

Draw a dotted horizontal line at $y = 5$ and read off the coordinates of the point on $y = 2x - 3$ crossed by this line.

This point is (4, 5), which means the solution to $2x - 3 = 5$ is $x = 4$.

Check: $2 \times 4 - 3 = 5$



EXERCISE 11.07 ANSWERS ON P. 649

Solving linear equations graphically U F

This exercise is best completed using graphing technology such as dynamic geometry software or graphing websites.

1 Use the graph above to solve each equation. Check your solutions.

a $2x - 3 = 3$ **b** $2x - 3 = -7$ **c** $2x - 3 = 0$

EXAMPLE 18

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2 Graph $y = 2x + 1$ and use it to solve each equation graphically.

a $2x + 1 = 7$ **b** $2x + 1 = 10$ **c** $2x + 1 = -5$

3 Solve each equation in question **2** algebraically.

4 At which point do the graphs of $y = 2x + 3$ and $y = 1$ intersect? Select the correct answer **A**, **B**, **C** or **D**.

A $(2, 1)$ **B** $(-1, 1)$ **C** $(4, -2)$ **D** $(8, -4)$

5 Graph $y = 3x - 2$ and use it to solve each equation graphically.

a $3x - 2 = 7$ **b** $3x - 2 = -11$ **c** $3x - 2 = -2$

6 Graph $y = -x + 1$ and use it to solve each equation graphically.

a $-x + 1 = -3$ **b** $-x + 1 = 4$ **c** $-x + 1 = -1$

7 Graph $y = -2x - 1$ and use it to solve each equation graphically.

a $-2x - 1 = -3$ **b** $-2x - 1 = 0$ **c** $-2x - 1 = 3$

8 Graph $y = \frac{1}{2}x + 3$ and use it to solve each equation graphically.

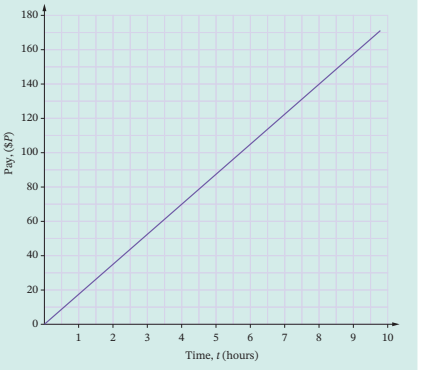
a $\frac{1}{2}x + 3 = 5$ **b** $\frac{1}{2}x + 3 = 1$ **c** $\frac{1}{2}x + 3 = 2$

11.08 Applying linear equations

Linear equations can be used to describe real-world situations.

Example 19

Jasmine is paid $17.50 per hour. The graph shows Jasmine's pay for a certain number of hours.



a What is the gradient of the line and what does it represent?

b Find the equation of the line.

c Use the graph to find Jasmine's pay for 8 hours.

d Use the graph to find the hours worked if Jasmine's pay was $105.

e Use the equation to find Jasmine's pay if she worked for:

i 3.5 hours **ii** 15 hours

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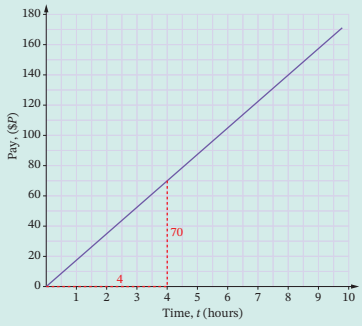
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Solution

a When $t = 4$, $P = 70$. Use this to find the gradient.

$$\text{Gradient} = \frac{70}{4}$$
$$= 17.5$$

The gradient represents Jasmine's rate of pay of \$17.50 per hour.



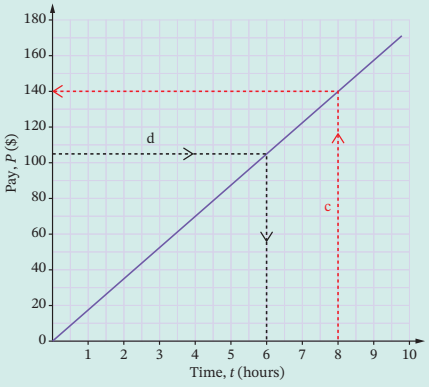
b Using $y = mx + c$ where $m = 17.5$, $c = 0$: $y = 17.5x$
But for this graph, the variables are P and t.
The equation is $P = 17.5t$.

c Reading from the graph, when $t = 8$, $P = 140$.
The pay for 8 hours is \$140.

d Reading from the graph, when $P = 105$, $t = 6$.
The pay of \$105 is for 6 hours' work.

e **i** $P = 17.5 \times 3.5$ $P = 17.5t$
$= \$61.25$
Pay of \$61.25 for 3.5 hours' work.

ii $P = 17.5 \times 15$
$= \$262.50$
Pay of \$262.50 for 15 hours' work.



Example 20

Sumaiyaa bought a computer for \$1500. Its value is depreciated each year using the graph shown.

a What was the value of the computer after 3 years?

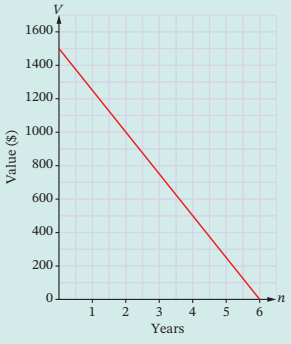
b Find the gradient and the vertical intercept of the line.

c Write down the equation of the line in terms of V and n.

d Use the equation to find the value of the computer after

i 4 years **ii** $5\frac{1}{2}$ years

e What is the amount of depreciation per year?

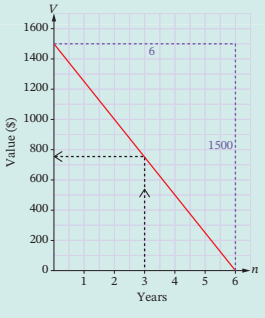


Solution

a Reading from the graph, the value of the computer after 3 years is \$750.

b $$\text{Gradient} = \frac{-1500}{6}$$
$$= -250$$

Vertical intercept = 1500



Vertical intercept is the general name for y-intercept when a variable other than y is used.

c $V = -250n + 1500$ $y = mx + c$

d **i** $n = 4, V = -250 \times 4 + 1500$
$= \$500$
Value after 4 years is \$500.

ii $n = 5\frac{1}{2}, V = -250 \times 5\frac{1}{2} + 1500$
$= \$125$
Value after $5\frac{1}{2}$ years is \$125.

e $m = -250$, so the depreciation is \$250 per year.

Video
Applying linear functions

Example 21

A taxi company charges a flagfall of \$2.90 and then charges \$2.17 per kilometre travelled.

Flagfall is the initial cost before any travelling starts.

a Write an equation to model the cost of a taxi trip for different distances.

b Construct a graph to model the costs for a trip from 0 to 50 km.

c Taksheel needs to use a taxi to get from his home to a friend's house 32 km away. Use your graph to determine the cost of this trip.

d Use the equation to determine the cost of this trip.

e Comment on any differences between your answers using the graph and equation.

Solution

a Let $y = mx + c$ where y is the cost of the taxi trip in dollars.

You could also name the equation $C = md + c$ where C = cost and d = distance.

The flagfall is a fixed amount and will be the starting value, so $c = 2.90$, the y-intercept.

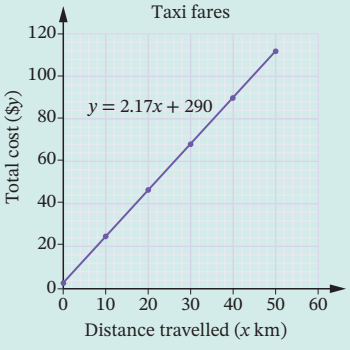
The charge per kilometre is a regular increase, so $m = 2.17$, which represents a positive gradient.

$y = 2.17x + 2.90$

b Construct a table of values that can be graphed. Distance goes up to 50 km so go up.

Spreadsheets and graphing software can also be used to complete the table and graph the equation.

x	0	10	20	30	40	50
y	2.90	24.60	46.30	68.00	89.70	111.40



c From the graph, when $x = 32$, $y = 72$.

The cost for travelling 32 km is approximately \$72.

d Algebraically, substitute $x = 32$ into the equation:

$y = 2.17x + 2.90$

$= 2.17 \times 32 + 2.9$

$= \$72.34$

e The algebraic solution is more accurate and 34 cents higher than the graphical solution. This is due to the scale used when constructing the graph.

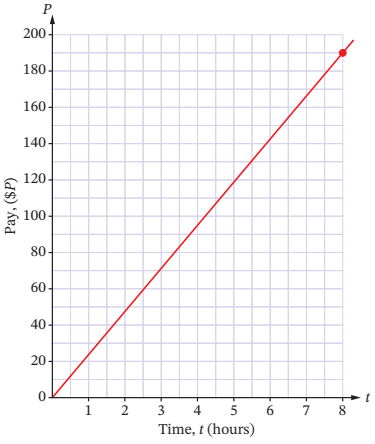
EXERCISE 11.08 ANSWERS ON P. 649

Applying linear equations U F R C

The questions in this exercise may also be completed with the aid of a spreadsheet and graphing software.

EXAMPLE 19

1 This graph shows Alex's pay for 8 hours' work. R C



a What is the gradient of the line and what does it represent?

b Find the equation of the line.

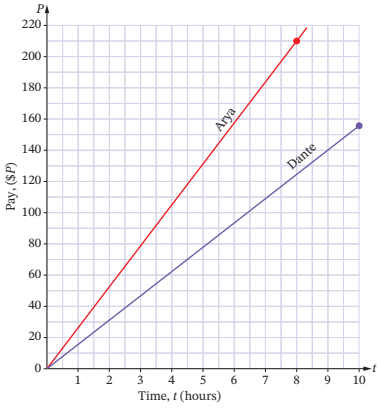
c Use the graph to find Alex's pay for 3.5 hours.

d Use the graph to find the hours worked if Alex's pay was \$142.50.

e Use the equation to find Alex's pay if he worked for:

i 7.5 hours **ii** 21 hours

2 This graph shows Arya's and Dante's pay. R C



a Find the gradient of each line.

b Find the equation of each line.

c Who earns more per hour? How is this shown by the graph of the lines?

d Use the graph to find the pay for Arya and Dante if they each worked for 4 hours.

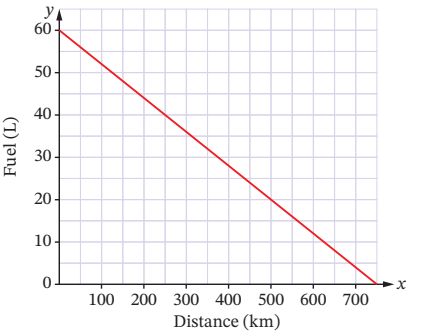
e Use the equations to find the pay for Arya and Dante if they worked for:

i 3.5 hours **ii** 10 hours **iii** 30 hours

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EXAMPLE 20

3 This graph shows the amount of fuel (y litres) remaining in a car after travelling a distance (x km) on a road trip. R C



a What is the y intercept and what does it represent?

b Find the gradient.

c Find the equation of the line.

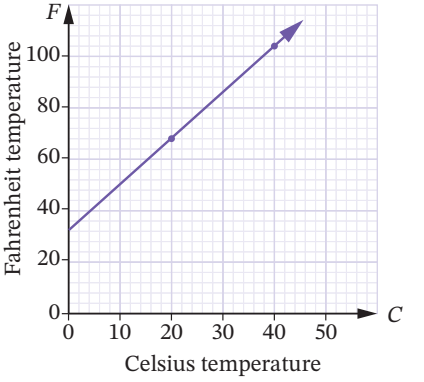
d How much fuel was remaining after 250 km?

e How much fuel was *used* after travelling 400 km?

f Use the equation to find the amount of fuel remaining after travelling:

i 300 km **ii** 700 km

4 The conversion of degrees Celsius (°C) temperatures to degrees Fahrenheit (°F) temperatures is shown by this graph. R C



a What is the temperature in °F when it is 0°C?

b Find the gradient of the line.

c Find the equation of the line (in the form $F = mC + c$).

d Use the equation to find the temperature in °F for:

i 25°C **ii** 34°C

e Use the graph to find the temperature in °C when it is 50°F.

f Use the equation to check your answer to part **e**.

EXAMPLE 21

5 An electrician charges a callout fee of $110 and then an hourly rate of $80. R C

a Copy and complete the table to show the electrician's charges.

Time, t (hours)	0	1	2	3	4
Cost ($C)	110	190			

b Graph this table of values.

c Find the gradient and vertical intercept of the graph.

d Find the linear equation of the graph.

e Use the graph to find the cost for 1.5 hours' work.

f Use the equation to find the cost for

i 2.5 hours **ii** 5.25 hours

g Use the equation to find the number of hours if the electrician charged $610.

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6 A school is organising a function for Year 10 students. The initial cost of catering will be $300 and then $10 per student. R C

a Copy and complete this table to show the cost of the function.

Number of students (x)	0	10	20	30	40	50	100	150
Cost ($\$y$)								

b Construct a graph that shows the cost of the function for 0 to 180 students. Use a scale of 1 cm = 20 students on the x-axis and 1 cm = $200 on the y-axis.

c What is the y-intercept of the graph and what does it represent?

d What is the gradient of the line and what does it represent?

e Find the linear equation in the form $y = mx + c$.

f Use the graph to find the cost of the function for:

i 90 students **ii** 145 students

g Use the equation to find the cost of the function for:

i 162 students **ii** 220 students.

h Find the number of students attending if the cost was $1650.

7 The equation $V = -750n + 4000$ describes the depreciated value of a photocopier, where V is the value in dollars and n is the time in years. R C

a Copy and complete this table of values for the equation.

Time (n years)	0	1	2	3	4	5
Value of photocopier ($\$V$)						

b Graph this table of values.

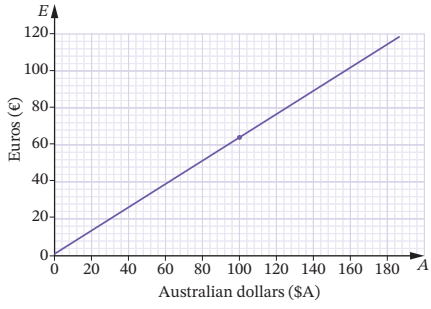
c What is the gradient of the line and what does it represent?

d What was the original value of the photocopier?

e Find the value of the photocopier after $3\frac{1}{2}$ years.

f Find the value of the photocopier after 6 years. What is the meaning of this result?

8 This graph can be used to convert Australian dollars ($A) to Euros (€). R C



a What is the vertical intercept?

b What is the gradient and what does it represent?

c Find the equation of the line (in the form $E = mA + c$)

d Use your equation to convert each amount to Euros.

i $A60 **ii** $A140 **iii** $A1500

e Use the graph to change each amount to Australian dollars.

i €80 **ii** €28 **iii** €120

9 Bradley has saved $120 so far for a holiday. He will add $15 to that amount each week for one year. R C

a Copy and complete this table of Bradley's savings.

Number of weeks (w)	0	1	2	3	4	5	6	7
Savings ($\$S$)								

b Graph this table of values.

c What is the vertical intercept of the line and what does it represent?

d Find the gradient of the line.

e Find the equation of the line in the form $S = mw + c$.

f How much will Bradley have in his savings after:

i 5 weeks? **ii** 12 weeks? **iii** 1 year?

g How long before Bradley has $360 in savings?

10 The graph shows the amount of water left in a leaking water tank. R C

a Find the gradient and vertical intercept of this line.

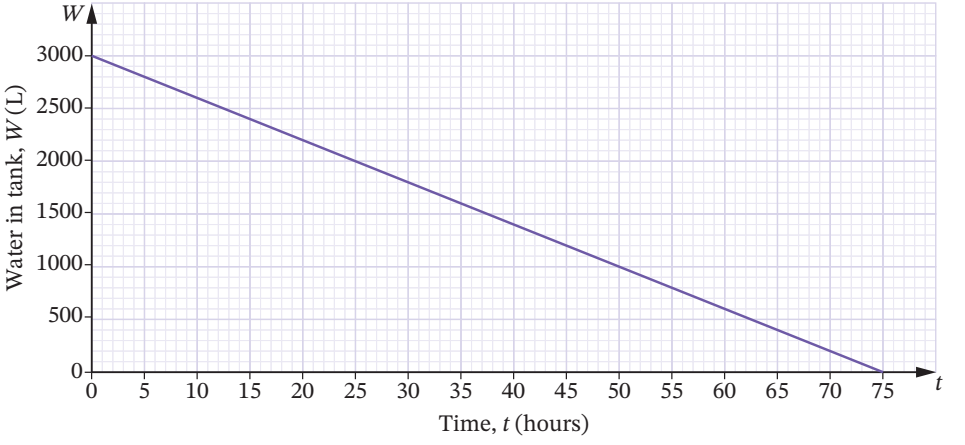
b Find the equation of the line.

c Find the amount of water in the tank after 45 hours using:

i the graph **ii** the equation

d Use the equation to find the amount of water in the tank after:

i 38 hours **ii** 3 days



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11 A sausage sizzle was organised at a high school fete. Cost of gas and other necessary items amounted to $45 and a sausage sandwich will cost $0.80 to make. R C

a Copy and complete the table to show the cost of making the sandwiches and then graph the table of values.

Number of sausage sandwiches (x)	0	20	40	60	80	100
Cost ($\$y$)						

b Find the linear equation of the line.

c Use the equation to find the cost of making 175 sandwiches.

d How many sausage sandwiches can be made for $245?

12 Christian borrowed $2300 from his parents to buy a mountain bike and has agreed to pay back $35 a week. R C

a Write an equation to show how much money ($\$A$) Christian owes after x weeks.

b Use the equation to find the amount owing after

i 40 weeks **ii** 1 year

Direct variation 11.09

PATH

Ann-Marie noticed a relationship between the amount of petrol used by her car and the distance travelled by the car, as shown in the table of values.

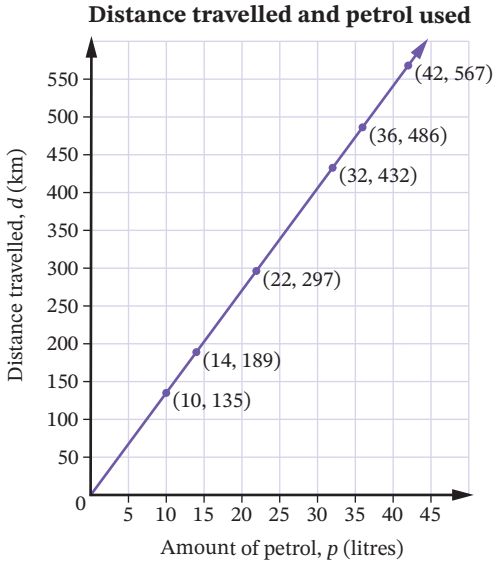
Amount of petrol used (L)	10	14	22	32	36	42
Distance travelled (km)	135	189	297	432	486	567

When she graphed these values, she found a linear relationship.

If p stands for the amount of petrol used and d stands for the distance travelled, then it can be shown that d is directly related to p by the formula $d = 13.5p$. This means that the distance travelled is a constant multiple of the amount of petrol used.

This is an example of **direct variation** (or **direct proportion**).

Two variables are **directly proportional** to each other if one variable is a constant multiple of the other, and when one variable changes, the other one changes by the same factor.



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Direct proportion

If y is **directly proportional** to x, written '$y \propto x$', then $y = kx$, where k is a constant (number) called the **constant of variation** or **constant of proportionality**.

- A direct linear relationship exists between x and y.
- If x increases (or decreases), y increases (or decreases).
- If x is doubled (or halved), y is doubled (or halved).
- Another way of saying 'y is directly proportional to x' is y **varies directly** as x'.
- A graph for direct variation is a straight line going through (0, 0) with gradient k.



Video
Direct proportion

Example 22

The distance (d), in metres, travelled by a car is directly proportional to the number of rotations (r) of its tyres. After 540 rotations, a distance of 950 m is travelled.

a What distance (to the nearest metre) will be travelled after 800 rotations?

b How many full rotations will be needed to cover 360 m?

Solution

a d is directly proportional to r.

$\therefore d = kr$ Forming a variation equation

To find k, substitute the information given for r and d.

When $r = 540$ and $d = 950$,

$950 = k \times 540$ Finding k

$k = \frac{950}{540}$

$= 1.759...$

$\therefore d = 1.759... r$ Rewriting the equation $d = kr$

When $r = 800$, Solving the problem

$d = 1.759... \times 800$

$= 1407.4074...$

≈ 1407 m

After 800 rotations, the distance travelled will be 1407 m.

b When $d = 360$, Solving the problem

$360 = 1.759... r$

$r = \frac{360}{1.759...}$

$= 204.661...$

≈ 205 rotations Rounding up for full rotations

For a distance of 360 m, there will be 205 rotations.

Solving direct variation problems

1 Identify the 2 variables (say x and y) and form a variation equation, $y = kx$.

2 Substitute values for x and y to find k, the constant of variation.

3 Rewrite $y = kx$ using the value of k.

4 Substitute a value for x or y into $y = kx$ to solve the problem.

PATH

11.09

EXAMPLE 22

EXERCISE 11.09 ANSWERS ON P. 651

Direct variation U F PS R

1 The stretch, S cm, of a spring is directly proportional to the mass, M kg, pulling on it.

Mass, M (kg)	0	8	12	20	24	40
Stretch, S (cm)	0	5	7.5	12.5	15	25

a Graph the linear relationship between S and M.

b The equation of the line is $S = kM$. Use the table of values to find the value of k as a decimal, then rewrite the equation for S.

c Use the equation to find the stretch caused by a mass of 36 kg.

d What is the mass that will cause a stretch of 40 cm?

2 B varies directly as h. If when $h = 5$, $B = 16$, what is the value of k in $B = kh$? Select the correct answer **A**, **B**, **C** or **D**.

A 80 **B** 11 **C** 3.2 **D** 0.3125

3 Y is directly proportional to X. If when $X = 15$, $Y = 75$, find the constant of variation, then use the variation equation to find Y when $X = 23$.

4 The mass, M, in grams of a chemical is directly proportional to its volume, V, in cm^3. PS R

a Write the formula $M = kV$, where k is a decimal, given that $M = 140$ when $V = 80$.

b Use the formula to calculate the mass of 406 cm^3 of the chemical.

c Calculate (correct to 1 decimal place) the volume of the chemical if the mass is:

i 28 g **ii** 0.36 kg

5 Find the linear formula for n in terms of m for this table of values.

m	3	6	9	12	15
n	7.5	15	22.5	30	37.5

6 p varies directly as q. If $q = 7$ when $p = 42$, find p when $q = 18$.

7 The distance, D, travelled by Pete, a marathon runner, varies directly as time, t. Pete runs 8.25 km in 44 minutes.

a How far will he run in one hour?

b How long will it take him to run 24 km?

PATH

8 For an object that is cooling, the drop in temperature varies directly with time.

If the temperature drops 8°C in 5 minutes, how long would it take to drop 10°C?
Select **A**, **B**, **C** or **D**.

A 6.25 min **B** 7 min **C** 12.8 min **D** 16 min

9 The amount of interest, I, earned on an investment account varies directly as the size of the deposit, D. PS R

a If Hannah earns $35 interest on an investment of $1400, write an equation relating I to D.

b How much will she earn on an investment of $2500?

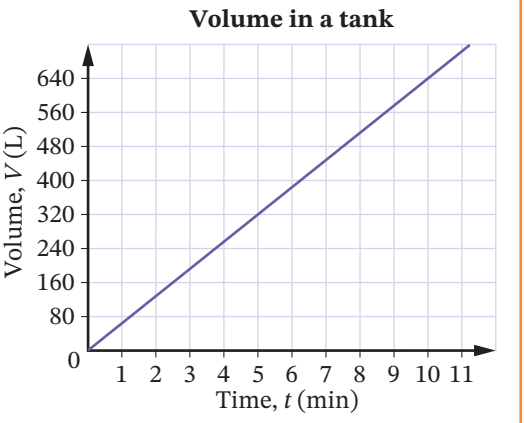
c If Hannah doubles the size of her investment in **b**, how much will she earn in interest?

10 The volume V (in litres) of water in a tank being filled is proportional to the time taken t (in minutes), as shown on the graph.

a Find the formula for V.

b Calculate the amount of water in the tank after 8 minutes.

c Calculate how long it takes to pump 960 L of water.



11 The weight of an astronaut on Mars is proportional to his weight on Earth. A 72 kg astronaut weighs 27.4 kg on Mars. PS R

a Calculate how much a 60 kg astronaut weighs on Mars, correct to 1 decimal place.

b If an astronaut weighs 32 kg on Mars, calculate his weight on Earth, correct to 1 decimal place.

12 The pressure experienced by a scuba diver is directly proportional to her depth under the water. The pressure at 25 m is 147 kilopascals (kPa). PS R

a What pressure is experienced at 40 m?

b At what depth is the pressure 833 kPa?

Investigation

Graphing $y = x^2$ and $y = ax^2$

You will need: 1 cm grid paper. This activity can also be completed using graphing software.

1 Copy and complete this table of values for $y = x^2$.

x	−4	−3	−2	−1	−0.5	0	0.5	1	2	3	4
y											

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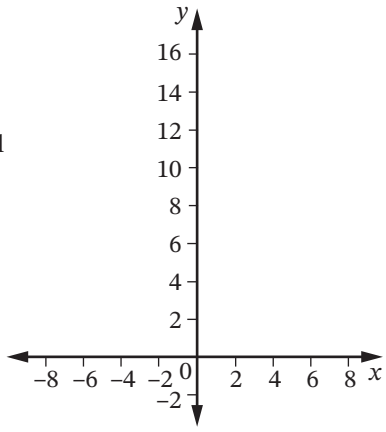
2 Draw this number plane on grid paper.

3 Plot the points from the table of values onto the number plane.

4 Draw a smooth curve through all the points and label the graph $y = x^2$.

5 Compare your graph with those of other students.

6 List as many features as you can about the graph of $y = x^2$ that you have drawn.
Compare your list with those of other students.



7 Copy and complete this table of values for $y = 2x^2$.

x	−3	−2	−1	0	1	2	3
y							

8 Graph $y = 2x^2$ on the same number plane as $y = x^2$ and label it.

9 How is the graph of $y = 2x^2$ different from the graph of $y = x^2$? Give an explanation for the difference.

10 On the same set of axes, draw these graphs:

a $y = x^2$ **b** $y = \frac{1}{2}x^2$ **c** $y = \frac{1}{3}x^2$

11 What do you notice about the graphs you have drawn? Give an explanation.

12 Compare your observations and explanation with those of other students.

Investigation

Graphing $y = ax^2 + c$

You will need: 1 cm grid paper. This activity can also be completed using graphing software.

1 Graph $y = x^2$ on a number plane for values of x from −3 to 3.

2 Copy and complete this table of values for $y = x^2 + 2$

x	−3	−2	−1	0	1	2	3
y							

3 Graph $y = x^2 + 2$ on the same number plane as $y = x^2$ and label it.

4 What do the graphs of $y = x^2$ and $y = x^2 + 2$ have in common?

5 How are the 2 graphs different?

6 Describe how the graph of $y = x^2 + 2$ could be drawn using the graph of $y = x^2$.

7 Graph each equation on the same number plane. Label your graphs clearly.

a $y = x^2 - 1$ **b** $y = x^2 + 4$ **c** $y = x^2 - 3$

8 Write a sentence to describe the effect of the constant term (which is added to, or subtracted from, x^2) on the shape and position of the parabola $y = x^2$.

9 Graph each equation on the same number plane. Label your graphs clearly.

a $y = -x^2$ **b** $y = -x^2 + 2$ **c** $y = -x^2 - 3$ **d** $y = -x^2 - 1$

10 What do you notice about the shape and position of each of the graphs in question **9**? Compare your results with those of other students.

11.10 Graphing quadratic equations $y = ax^2 + c$

In Chapter 7, *Equations*, we learnt that an equation in which the highest power of the variable is 2 is called a **quadratic equation**, for example, $y = 2x^2 - 5$, $y = x^2 + 7x + 12$ and $y = -5x^2$.

Worksheet Graphing parabolas

Technology Curve sketcher

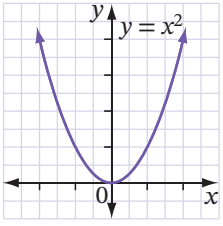
Spreadsheet Curve sketcher

Interactive Graphing quadratics

Video The parabola $y = ax^2 + c$

The graph of $y = x^2$

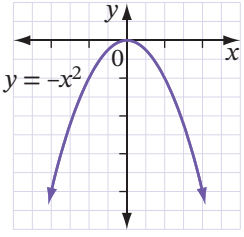
The graph of $y = x^2$ is a smooth U-shaped curve called a **parabola**.



- The axis of symmetry, called the **axis of the parabola**, is the y-axis.
- The **vertex** or **turning point** is (0, 0).
- Because x^2 is always positive or zero, the minimum value of the parabola is 0.
- The parabola's shape is **concave up** (looks like a smile ☺).

The graph of $y = -x^2$

The graph of $y = -x^2$ is the parabola $y = x^2$ reflected in the x-axis, because $-x^2$ is the 'negative' of x^2.

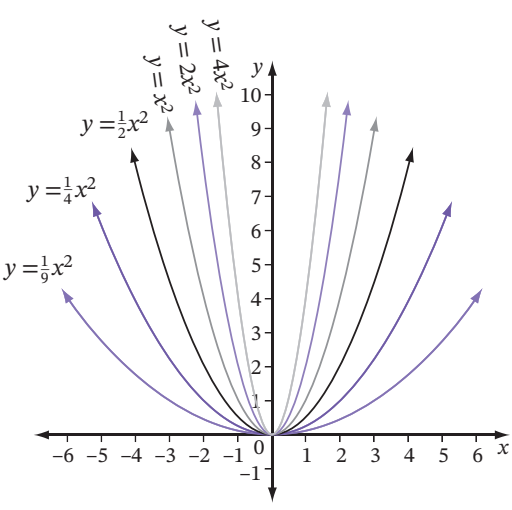


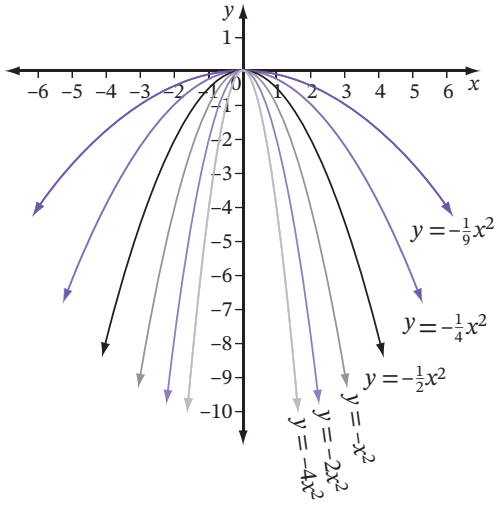
- The **axis of the parabola** is the y-axis.
- The **vertex** or **turning point** is (0, 0).
- Because $-x^2$ is always negative or zero, the maximum value of the parabola is 0.
- The parabola's shape is **concave down** (looks like a frown ☹).

The graph of $y = ax^2$

For the graph of a quadratic equation in the form $y = ax^2$, where a is a constant (number), the size of a affects whether the parabola is 'wide' or 'narrow'.

As the size of a increases, the parabola becomes 'narrower' and as the size of a decreases, the parabola 'widens'. If a is negative, then the parabola is concave down.





11.10

Example 23

a Graph $y = -x^2$ and $y = -2x^2$ on the same set of axes.

b What is the vertex of each parabola?

c Are the parabolas concave up or concave down?

d At what point do the 2 parabolas meet?

e Which parabola is 'wider'?

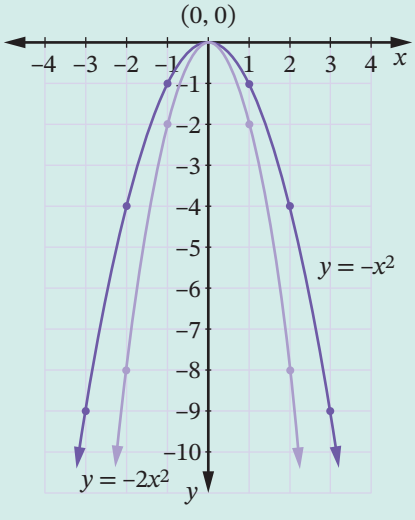
Solution

a $y = -x^2$

x	−3	−2	−1	0	1	2	3
y	−9	−4	−1	0	−1	−4	−9

$y = -2x^2$

x	−3	−2	−1	0	1	2	3
y	−18	−8	−2	0	−2	−8	−18



b The vertex of both parabolas is (0, 0).

c The parabolas are both concave down.

d The parabolas meet at (0, 0).

e The parabola $y = -x^2$ is wider.

The graph of $y = ax^2 + c$

For the graph of a quadratic equation in the form $y = ax^2 + c$, where a and c are constants, the effect of c is to move the parabola up or down from the **origin**. Also, c is the y-intercept of the parabola.

Example 24

a Graph $y = x^2$, $y = x^2 + 2$, $y = -x^2 - 1$ and $y = -x^2 - 4$ on the same set of axes.

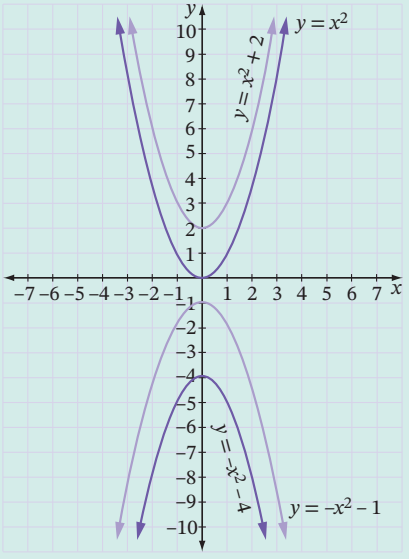
b Compare each parabola with the parabola $y = x^2$.

c What is the vertex of each parabola?

d Is each parabola concave up or concave down?

Solution

a



b $y = x^2 + 2$ is $y = x^2$ shifted up 2 units.

$y = -x^2 - 1$ is $y = x^2$ turned upside down and shifted down 1 unit.

$y = -x^2 - 4$ is $y = x^2$ turned upside down and shifted down 4 units.

c $y = x^2$ has vertex $(0, 0)$.

$y = x^2 + 2$ has vertex $(0, 2)$.

$y = -x^2 - 1$ has vertex $(0, -1)$.

$y = -x^2 - 4$ has vertex $(0, -4)$.

d $y = x^2$ and $y = x^2 + 2$ are concave up.

$y = -x^2 - 1$ and $y = -x^2 - 4$ are concave down.

EXERCISE 11.10 ANSWERS ON P. 652

Graphing quadratic equations $y = ax^2 + c$ U F R C

The questions in this exercise may also be completed using graphing software.

EXAMPLE 23

1 a Graph $y = 2x^2$ on a number plane after copying and completing this table of values.

x	−2	−1	0	1	2
y					

b Graph $y = \frac{1}{2}x^2$ on the same number plane after copying and completing this table of values.

x	−2	−1	0	1	2
y					

c Compare both parabolas. What features are the same? What is different? R C

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2 Which statement is correct about the parabola $y = x^2$? Select **A**, **B**, **C** or **D**.

A It has a y-intercept of 1.
B It is symmetrical about the x-axis.
C It is concave down.
D Its vertex is $(0, 0)$.

3 **a** Graph $y = -3x^2$ and $y = -0.1x^2$ on the same number plane after copying and completing this table of values for each equation. R C

x	−2	−1	0	1	2
y					

b Compare both parabolas. What features are the same? What is different?

4 Graph each pair of parabolas on the same set of axes and state: R C

i the vertex of each parabola
ii whether each parabola is concave up or concave down
iii which parabola is 'wider'
iv the point at which the 2 parabolas meet.

a $y = x^2$ and $y = \frac{1}{4}x^2$
b $y = 2x^2$ and $y = -2x^2$
c $y = -3x^2$ and $y = -\frac{1}{2}x^2$
d $y = 5x^2$ and $y = 0.2x^2$

5 Which statement is correct about the parabola $y = -6x^2$? Select **A**, **B**, **C** or **D**.

A It has a y-intercept of −6.
B It passes through $(-1, 6)$.
C It is concave down.
D Its vertex is $(0, -6)$.

6 Graph each pair of parabolas on the same set of axes and state: R C

i the vertex of each parabola
ii whether each parabola is concave up or concave down
iii which parabola is 'wider'
iv the y-intercept of each parabola.

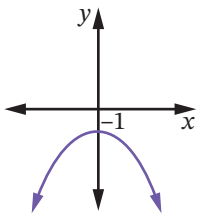
a $y = x^2 + 1$ and $y = x^2 + 3$
b $y = -x^2 - 2$ and $y = 2x^2 + 4$
c $y = -5x^2 - 2$ and $y = -2x^2 - 1$
d $y = -3x^2 - 4$ and $y = 2x^2 + 2$

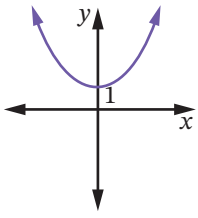
7 Which statement is correct about the parabola for $y = x^2 + 2$? Select **A**, **B**, **C** or **D**.

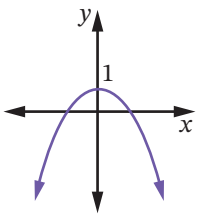
A It has a maximum value of 2.
B It is symmetrical about the x-axis.
C It is concave up.
D Its vertex is $(2, 0)$.

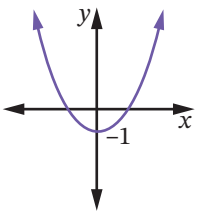
8 Match each quadratic equation to its parabola.

a $y = x^2 - 1$ **b** $y = x^2 + 1$ **c** $y = -x^2 - 1$ **d** $y = -x^2 + 1$

A


B


C


D


11.10

EXAMPLE 24

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11.11 Graphing quadratic equations $y = ax^2 + bx + c$

PATH

Worksheets
Graphing quadratic functions

Graphing quadratics

Features of a parabola

Now we will graph parabolas of the general form $y = ax^2 + bx + c$.

Example 25

For each quadratic equation:

i complete a table of values and draw the graph

ii find the x-intercepts

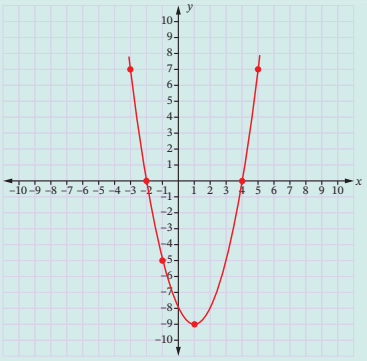
iii find the y-intercept

a $y = x^2 - 2x - 8$ **b** $y = 2x^2 + x - 6$

Solution

a **i** $y = x^2 - 2x - 8$

x	−3	−2	−1	0	1	2	3	4	5
y	7	0	−5	−8	−9	−8	−5	0	7



ii The x-intercepts are −2 and 4.

From the graph and table.

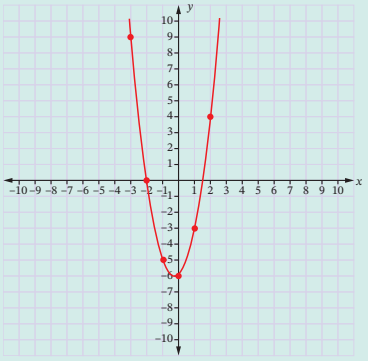
iii The y-intercept is −8.

From the graph and table.

Note that this is the constant term $c = -8$ in $y = x^2 - 2x - 8$.

b **i** $y = 2x^2 + x - 6$

x	−3	−2	−1	0	1	2
y	9	0	−5	−6	−3	4



ii The x-intercepts are −2 and $1\frac{1}{2}$.

iii The y-intercept is −6.

Note that this is the constant term $c = -6$ in $y = 2x^2 + x - 6$. Also note that (0, −6) is NOT the vertex. The vertex is slightly left and below it at (−0.25, −6.125).

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The graph of the parabola $y = ax^2 + bx + c$

If $a > 0$ (positive), the parabola is **concave up**

If $a < 0$ (negative), the parabola is **concave down**

The **y-intercept** is where $x = 0$, so it is the value of c, the constant term

The **x-intercepts** are where $y = 0$

Because the parabola is symmetrical, the x-coordinate of its vertex is halfway between the x-intercepts.

Example 26

For each quadratic equation:

i find the x-intercepts of its graph

ii find the y-intercept

iii find the coordinates of the vertex

iv draw its graph

a $y = x^2 + x - 12$ **b** $y = x(2x - 9)$

Solution

a **i** Substitute $y = 0$ to find x-intercepts:

$0 = x^2 + x - 12$

$x^2 + x - 12 = 0$

$(x + 4)(x - 3) = 0$

$x + 4 = 0$ or $x - 3 = 0$

$x = -4$ or $x = 3$

The x-intercepts are -4 and 3.

ii Look at the constant term, $c = -12$, or substitute $x = 0$ to find the y-intercept.

The y-intercept is -12.

iii By symmetry, the x-coordinate of the vertex is halfway between the x-intercepts -4 and 3.

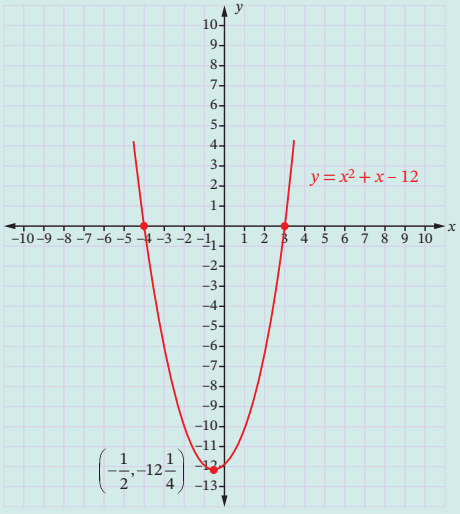
$$x = \frac{-4 + 3}{2} = -\frac{1}{2}$$

Substitute $x = -\frac{1}{2}$ to find y:

$$y = \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right) - 12 = -12\frac{1}{4}$$

Vertex $\left(-\frac{1}{2}, -12\frac{1}{4}\right)$

iv Draw the graph.



PATH

b i Substitute $y = 0$ to find x-intercepts:

$0 = x(2x - 9)$

$x(2x - 9) = 0$

$x = 0$ or $2x - 9 = 0$

$2x = 9$

$x = 4\frac{1}{2}$

The x-intercepts are 0 and $4\frac{1}{2}$.

ii Substitute $x = 0$ to find the y-intercept:

$y = 0(2(0) - 9)$

$= 0 \times (-9)$

$= 0$

The y-intercept is 0.

iii By symmetry, the x-coordinate of the vertex is halfway between the x-intercepts 0 and $4\frac{1}{2}$.

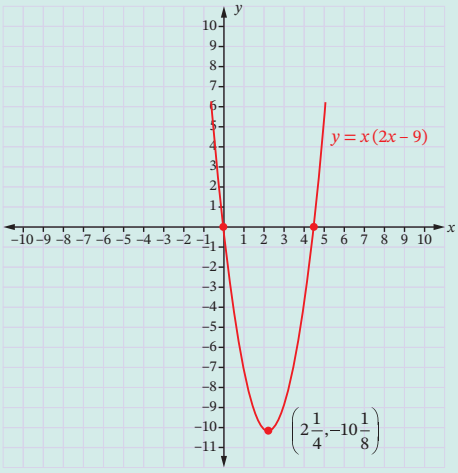
$$x = \frac{0 + 4\frac{1}{2}}{2} = 2\frac{1}{4}$$

Substitute $x = 2\frac{1}{4}$ to find y:

$$y = 2\frac{1}{4}\left[2\left(2\frac{1}{4}\right) - 9\right] = -10\frac{1}{8}$$

Vertex $\left(2\frac{1}{4}, -10\frac{1}{8}\right)$

iv Draw the graph.



EXERCISE 11.11 ANSWERS ON P. 653

Graphing quadratic equations $y = ax^2 + bx + c$ UFRC

EXAMPLE 25

1 For each quadratic equation:

i complete a table of values and draw the graph

ii find the x-intercepts

iii find the y-intercept

a $y = x^2 + 5x + 4$ **b** $y = 2x^2 - 9x + 7$

c $y = x^2 - 6x$ **d** $y = 2x^2 + 11x + 9$

EXAMPLE 26

2 Find the y-intercept of the parabola with equation:

a $y = 3x^2 + 5x - 4$ **b** $y = 2x^2 - 5x + 11$ **c** $y = 8x^2 + x$

3 Graph each equation, showing its x- and y-intercepts and vertex.

a $y = (x + 2)(x - 3)$ **b** $y = x^2 - 7x + 6$ **c** $y = x(x - 3)$

d $y = 4x(x + 2)$ **e** $y = -x^2 - 3x + 40$ **f** $y = x^2 + 12x + 35$

g $y = (x + 4)(2x - 5)$ **h** $y = (2x + 3)(2x - 5)$ **i** $y = -x^2 + 10x - 25$

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4 Which quadratic equation below is a parabola with x-intercepts 4 and −7? Select **A**, **B**, **C** or **D**.

A $y = x^2 - 3x - 28$ **B** $y = x^2 + 3x - 28$
C $y = x^2 - 3x + 28$ **D** $y = x^2 + 3x + 28$

5 Which quadratic equation below is a parabola with only one x-intercept of 2 and a y-intercept of 20? Select **A**, **B**, **C** or **D**.

A $y = x^2 - 2x - 5$ **B** $y = (x - 2)(x - 5)$
C $y = (x - 2)(x - 10)$ **D** $y = 5(x - 2)^2$

PATH

Applying quadratic equations 11.12

Parabolas can be seen in our everyday lives. When throwing a stone or kicking a ball, the path of the object is a parabola.

Video
Applying quadratic functions

Example 27

A ball is dropped from a balcony and its height (y metres) at any time (t seconds) is given by $y = -5t^2 + 45$.

a Complete this table of values to graph the equation.

Time (t seconds)	0	1	2	3	4
Height (y metres)					

b What is the y-intercept and what does this value represent?

c Use the graph to find the height of the ball after 1.5 seconds.

d How far has the ball dropped after 2 seconds?

e Find the total time the ball is in the air.

Solution

a

Time (t seconds)	0	1	2	3	4
Height (y metres)	45	40	25	0	−35

Spreadsheets and graphing software can also be used to complete the table and graph the equation.



We can ignore (4, −35) because height can't be negative and the ball has reached the ground at $t = 3$.

b y-intercept is 45, which represents the height of the balcony in metres

c 34 m

d Height after 2 s is 25 m, so the ball has dropped 20 m.

e 3 seconds (when the height of the ball is 0)

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EXERCISE 11.12 ANSWERS ON P. 655

Applying quadratic equations U F R C

The questions in this exercise may also be completed with the aid of a spreadsheet and graphing software.

EXAMPLE 27

1 A stone is dropped from a 50 m cliff and its height above ground level (h metres) at time t seconds is given by the quadratic equation $h = 50 - 4.9t^2$. R

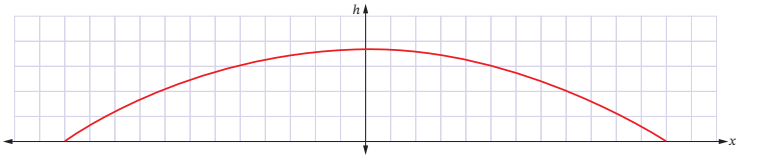
a Copy and complete this table of values to graph the equation.

Time, t (seconds)	0	1	2	3	4
Height, h (metres)					

b What is the height of the ball after 2 seconds?

c Estimate when the stone will hit the ground.

2 A bridge has a parabolic span as shown with equation $h = -0.005x^2 + 18$ where h is the height of the bridge in metres and x is the horizontal distance from the centre of the span in metres. R C



a Complete a table of values for x values from –60 to 60 (using intervals of 10), then graph the equation.

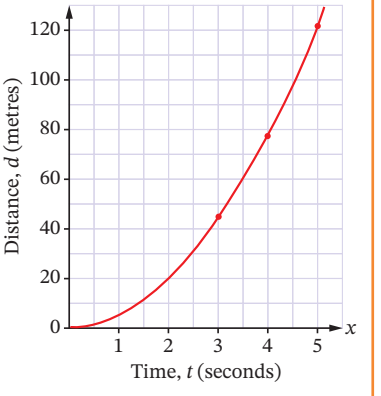
b Find the height of the bridge at its highest point.

c Find the total width of the span.

d What is the height of the span when the width is 80 m?

3 A ball is dropped from the top of a building 80 m high and the distance, d metres, it falls after t seconds is given by the equation $d = 4.9t^2$, also graphed here. R C

t	0	1	2	3	4	5
d	0	4.9	19.6	44.1	78.4	122.5



a How far has the ball fallen after:

i 2 seconds?

ii 3 seconds?

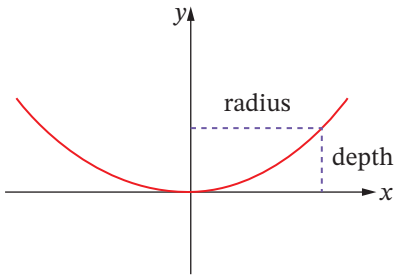
b How far did the ball fall:

i in the first second?

ii in the third second?

c After how many seconds did the ball hit the ground?

4 The inside of a satellite dish is the shape of a parabola and can be modelled by the equation $y = 0.04x^2$, where y metres is the depth and x metres is the radius.



a Graph $y = 0.04x^2$ for values of x from -6 to 6.

b Find the radius of the dish where the depth is 1 m.

c Find the depth of the dish where the radius is:

i 3.5 m **ii** 4 m

5 The shape of the cables between the 2 towers of the main span of the Golden Gate Bridge in San Francisco is very close to a parabola. The parabola is modelled by the equation $h = \frac{19}{51\,200}x^2 + 75$, where h is the height of the cables above the water level. R C



Shutterstock.com/bgrissom

The shape is actually a special curve called a **catenary**, caused by gravity acting on the hanging cables.

a Graph the equation for x values between -700 and 700, going up in steps of 100.

b What is the height above the water level of the cables at their lowest point?

c Use the graph to find the values of x where the height of the cable is 150 m.

d If the height of the towers at either end of the span is 227 m, estimate the distance between the towers.

e If $x = 250$, find the height of the cables above the water level.

6 The height y metres of a golf ball after time x seconds is given by the equation

$$y = -0.9375x^2 + 7.5x.$$ R C

a Graph this equation for x values from 0 to 8.

b What is the maximum height the ball reaches and when does it reach this height?

c At what times does the golf ball reach a height 14 m? Explain why there are 2 answers.

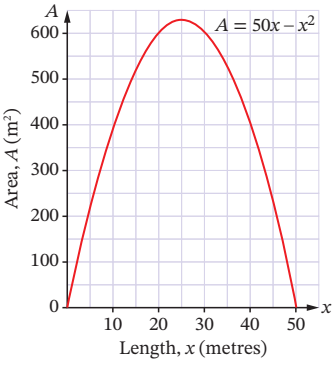
7 The area of a rectangular block of land is given by the equation $A = 50x - x^2$, where x metres is the length of the block. The equation is graphed below. R C

a What is the maximum area of the block of land and what is the length?

b Find the width of the block of land with maximum area.

c Use the graph to find the length and width of the block if the area is:

i 600 m^2 **ii** 400 m^2.



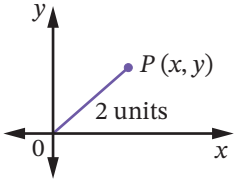
11.13 Graphing circles $x^2 + y^2 = r^2$

PATH

Technology
Curve sketcher

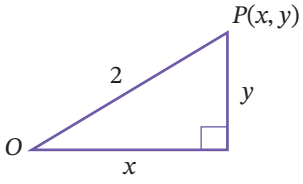
Spreadsheet
Curve sketcher

A circle is a completely round shape. Every point on a circle is exactly the same distance from its centre. We can use this fact to find the equation of a circle with a radius of 2 units.



Let $P(x, y)$ be any point so that the distance $OP = 2$ units and O is the origin.

If we plotted every possible position of P, we would graph a circle centred at O with a radius of 2. We can use Pythagoras' theorem to find the equation of this circle by drawing a right-angled triangle with OP as the hypotenuse.



Since P has coordinates (x, y), the triangle must have a base length of x and a height of y, so by Pythagoras' theorem:

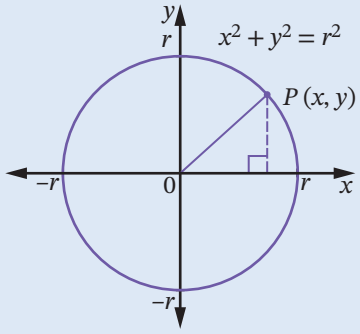
$x^2 + y^2 = 2^2 = 4$

$\therefore$ The equation of a circle with centre (0, 0) and radius 2 is $x^2 + y^2 = 4$.

This can be generalised for a circle of any radius.

Equation of a circle with centre (0, 0)

The **equation of a circle** with centre (0, 0) and radius r units is $x^2 + y^2 = r^2$.



We can use this formula to find the centre and radius of any circle.

Example 28

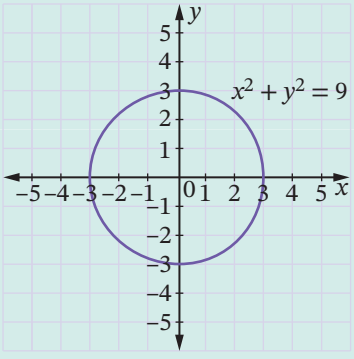
Find the centre and radius of the circle $x^2 + y^2 = 9$ and sketch its graph.

Solution

The centre is (0, 0). The radius is r, where $r^2 = 9$.

$r = \sqrt{9}$

$\therefore$ Radius, $r = 3$ units



Example 29

Find the equation of this circle.

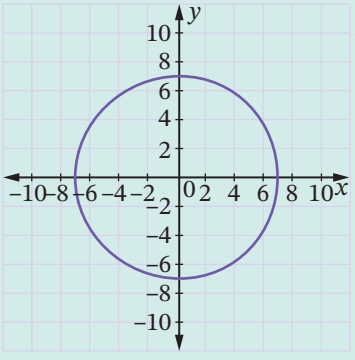
Solution

The circle has a radius of 7 units and the centre is (0, 0).

The equation of the circle is

$x^2 + y^2 = 7^2$

$x^2 + y^2 = 49$



PATH

EXERCISE 11.13 ANSWERS ON P. 657

Graphing circles U F

The questions in this exercise may also be completed using graphing software.

EXAMPLE 28

1 Find the centre and radius of the circle given by each equation and sketch its graph.

a $x^2 + y^2 = 4$ **b** $x^2 + y^2 = 1$ **c** $x^2 + y^2 = 81$

d $x^2 + y^2 = 100$ **e** $x^2 + y^2 = 25$ **f** $x^2 + y^2 = \frac{1}{4}$

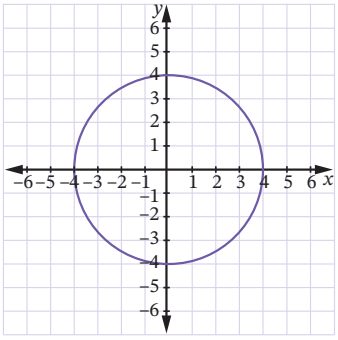
2 What is the equation of a circle with centre (0, 0) and radius 6 units?
Select the correct answer **A**, **B**, **C** or **D**.

A $x^2 + y^2 = 12$ **B** $x^2 + y^2 = 6$ **C** $x^2 + y^2 = 8$ **D** $x^2 + y^2 = 36$

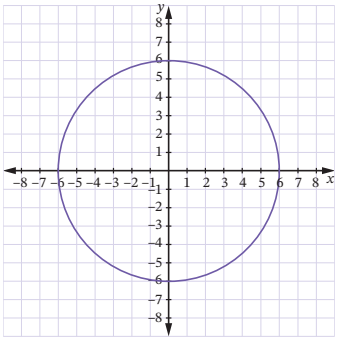
EXAMPLE 29

3 Find the equation of each circle shown.

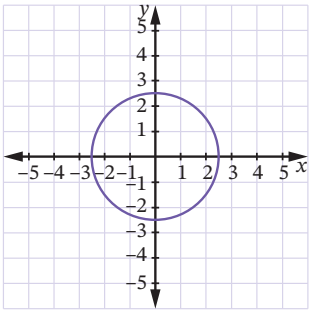
a



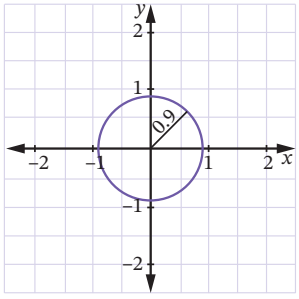
b



c



d



4 What is the equation of a circle with centre (0, 0) and radius 1.3 units? Select **A**, **B**, **C** or **D**.

A $x^2 + y^2 = 2.6$ **B** $x^2 + y^2 = 1.69$ **C** $x^2 + y^2 = 1.72$ **D** $x^2 + y^2 = 3.69$

5 What is the radius of a circle with equation $x^2 + y^2 = 144$? Select **A**, **B**, **C** or **D**.

A 12 units **B** 14 units **C** 36 units **D** 72 units

6 What is the equation of a circle with centre (0, 0) and diameter 16 units?

7 What is the radius of a circle with equation $x^2 + y^2 = \frac{9}{4}$? Select **A**, **B**, **C** or **D**.

A $2\frac{1}{4}$ units **B** $1\frac{1}{2}$ units **C** $\frac{3}{4}$ units **D** $\frac{7}{8}$ units

8 What is the diameter of the circle whose equation is $x^2 + y^2 = 196$?

9 **a** What is the radius of the circle whose equation is $x^2 + y^2 = 12$?

b Sketch the graph of $x^2 + y^2 = 12$

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Power plus ANSWERS ON P. 657

Puzzle
Finding coordinates for given segment lengths

11.13

1 $ABCD$ is a parallelogram. The coordinates of A, B and C are $(-1, 4)$, $(4, 6)$ and $(2, 7)$ respectively. Find the coordinates of D.

2 A line drawn through points $A(0, -2)$ and $B(3, 0)$ also goes through point C with x-coordinate 9. Find:

a the equation of the line

b the y-coordinate of C.

3 $M(1, 1)$ is the midpoint of the interval XY. If the coordinates of X are $(-3, -2)$, what are the coordinates of Y?

4 A circle is drawn with its centre at $C(2, 3)$. The point $A(-1, -5)$ lies on the circumference.

a Calculate the radius of the circle.

b The point P is 10 units from C. Explain why P lies outside the circle.

5 The points of $R(2, 19)$, $S(-3, 7)$ and $T(10, 7)$ are joined to form a triangle. Show that the triangle is isosceles.

6 The point $(2, 1)$ is a vertex of a square. $M(5, 1)$ is the point of intersection between the diagonals of the square. Find the coordinates of the other 3 vertices of the square.

7 Three vertices of a quadrilateral are $(-3, -3)$, $(4, -1)$ and $(6, 4)$. Find the coordinates of the fourth vertex so that the quadrilateral is parallelogram.

8 P is proportional to the *square* of M. If $P = 400$ when $M = 16$, find:

a P when $M = 20$ **b** M when $P = 256$

CHAPTER 11 REVIEW

Quiz
Language of maths 11

Puzzle
Coordinate geometry crossword

Language of maths

axis/axes	constant of variation	direct variation
distance	exact	formula
gradient	horizontal	interval
length	linear equation	midpoint
number plane	parabola	quadratic equation
rise	run	surd
varies directly	vertex	vertical
vertical intercept	x–intercept	y–intercept

1 What is the name given to the graph of:

a a quadratic equation? **b** a linear equation?

2 Is the line $y = 3$ horizontal or vertical?

3 What is another name for the steepness of a line?

4 What is the value at which a line crosses the y-axis called?

5 What are the coordinates of the **vertex** of the parabola with equation $y = -x^2$?

6 If $y = kx$, what is the constant of variation?

7 When the length of an interval needs to be given as an **exact** length, what does this mean?

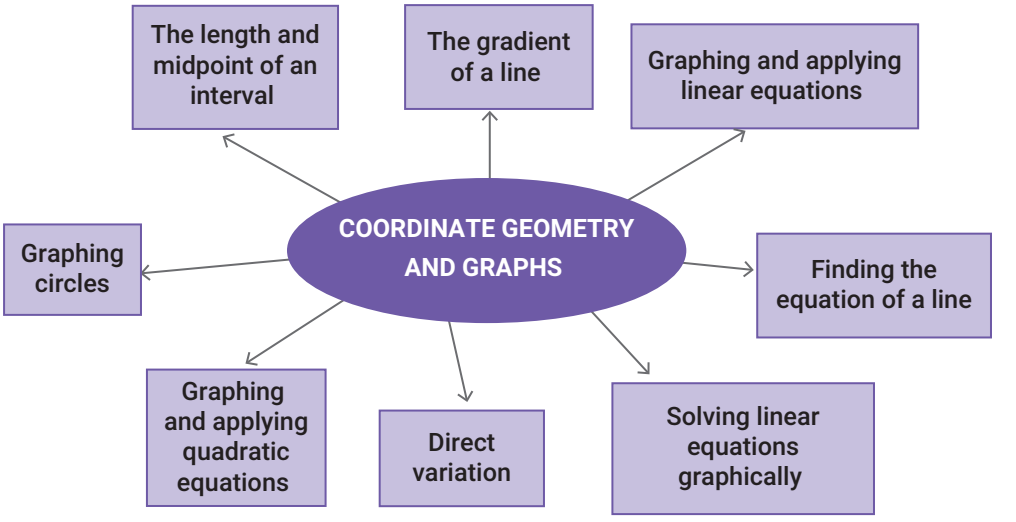
8 What is the equation of the x-axis?

Topic summary

Worksheet
Mind map: Coordinate geometry and graphs

- In your own words, describe what a gradient is.
- What parts of this topic did you find difficult?

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 11 ANSWERS ON P. 657

Quiz
Test yourself 11

1 Find the length of the interval joining the points $H(7, 2)$ and $T(-2, 6)$. Express your answer: 11.01

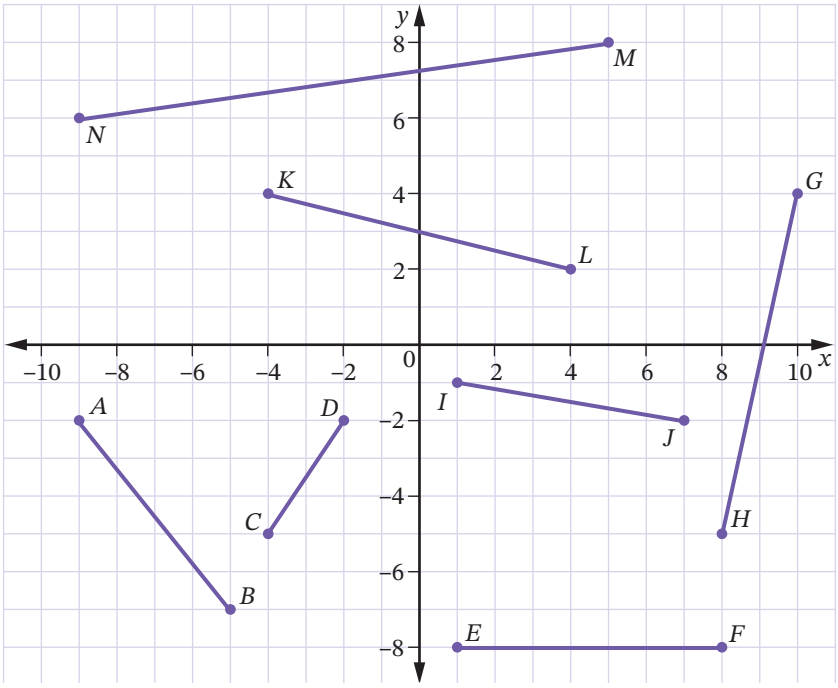
a as a surd **b** correct to one decimal place.

2 Find the coordinates of the midpoint of the interval joining: 11.02

a (8, 4) and (−2, 10) **b** (−1, −1) and (−7, 9) **c** (3, −6) and (−5, 10)

3 **a** Which intervals below have a gradient that is: 11.03

i positive? **ii** negative? **iii** 0?



b Find the gradient of:

i interval NM **ii** interval AB **iii** interval KL

c Find the midpoint of interval HG.

d Find the exact length of interval CD.

4 Find the gradient of the interval joining: 11.03

a (1, 9) and (3, 5) **b** (6, 0) and (8, −4) **c** (−1, −1) and (5, −3)

5 Graph $y = -3x + 6$ on a number plane and write: 11.04

a its x-intercept **b** its y-intercept.

6 Test whether the point (4, −2) lies on the line with equation: 11.04

a $y = x - 6$ **b** $y = -2x + 6$ **c** $x - 3y = 10$

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11.04 **7** **a** Graph and label each line on the same number plane:

$x = -4, \quad x = 3, \quad y = -6, \quad y = 2$

b What is the area of the rectangle enclosed by these lines?

11.04 **8** Write the equation of a line that is:

a horizontal with a y-intercept of 4

b parallel to the y-axis and passes through $(-1, 7)$

c the x-axis

11.05 **9** Find the gradient and y-intercept of the line with equation:

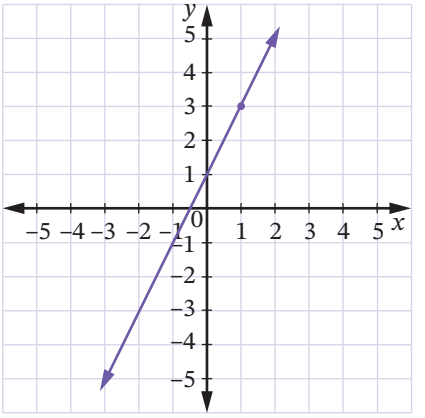
a $y = 3x + 1$ **b** $y = 8 - x$ **c** $y = \frac{x}{5} + 4$ **d** $y = -6x$

11.05 **10** Graph each linear equation by finding the gradient and y-intercept first.

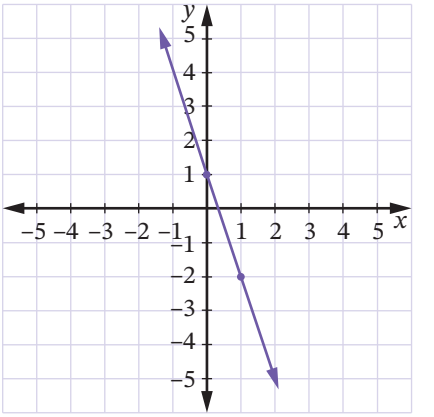
a $y = -3x - 1$ **b** $y = \frac{x}{2} + 6$

11.06 **11** Find the equation of each line.

a



b



11.07 **12** Solve the equation $3x - 4 = 5$ graphically.

13 The cost of using a ridesharing service is a flagfall of \$3.50 and then a charge of \$2.25 per kilometre travelled.

11.08 **a** Copy and complete this table.

Distance (x km)	0	1	2	3	4	5	6
Cost (\$$y$)							

b Graph the table of values.

c Use the graph to find the cost of a 3.5 km trip.

d Find the gradient and y-intercept of the line.

e Find the equation of the line.

f Use the equation to find the cost of the taxi for a distance of:

i 12 km **ii** 76 km

g Find the distance travelled if the cost was:

i \$12.05 **ii** \$82.25

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PATH

14 During a thunderstorm, the time, T seconds, between seeing lightning and hearing thunder varies directly as the distance, d kilometres, of the observer from the centre of the storm. Thunder is heard 6 seconds after the lightning is seen when a storm is 35 km away. 11.09

a Find a formula for T in terms of d.

b Calculate, correct to the nearest kilometre, the distance of a storm when thunder is heard 10 seconds after the lightning.

15 Graph $y = x^2 + 1$ and $y = -2x^2$ on the same set of axes and state: 11.10

a the vertex of each parabola

b whether each parabola is concave up or concave down

c which parabola is 'wider'.

16 Which equation below represents a concave up parabola with vertex (0, 0)? Select **A**, **B**, **C** or **D**. 11.10

A $y = x^2 - 1$ **B** $y = 2x^2$ **C** $y = -x^2$ **D** $y = 3x^2 + 2$

17 A stone is dropped from a cliff and its height y metres at time t seconds is given by $y = -5t^2 + 20$. 11.12

a Copy and complete this table of values and graph the equation.

Time (t seconds)	0	1	2	3
Height (y metres)				

b What is the y-intercept and what does it represent?

c What is the height of the stone after:

i 0.5 seconds? **ii** 1.5 seconds?

d When will the height of the stone be 5 m?

e When will the stone hit the ground?

f What distance does the stone drop in the first second?

PATH

18 Graph each quadratic equation, showing x- and y-intercepts and the vertex. 11.11

a $y = -x^2 + x + 12$ **b** $y = 3x^2 - 9x$

19 Find the centre and radius of the circle with equation $x^2 + y^2 = 16$ and sketch its graph. 11.13

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